

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.1	6.5	1.6	0.5	0	77.3
U2 - Beta(0.5, 0.5)	14.7	7.2	1	0.2	0	76.9
U2 - Beta(0.7, 0.3)	14.9	6.1	1.4	0.2	0.1	77.3
U2 - Beta(1, 1)	15.1	6.7	1	0.1	0	77.1
U3 - Beta(0.3, 0.7)	15.7	6.4	0.9	0.1	0.1	76.8
U3 - Beta(0.5, 0.5)	15.8	6.2	1.1	0.1	0	76.8
U3 - Beta(0.7, 0.3)	16.3	6	0.9	0.1	0	76.7
U3 - Beta(1, 1)	17.2	5	0.8	0	0	77
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	3.3	1.7	0.7	0.2	-
U2 - Beta(0.5, 0.5)	11.3	3.3	1.7	0.6	0.2	-
U2 - Beta(0.7, 0.3)	11.4	3.3	1.6	0.6	0.1	-
U2 - Beta(1, 1)	11.3	3.3	1.7	0.6	0.2	-
U3 - Beta(0.3, 0.7)	11.3	3.3	1.7	0.7	0.2	-
U3 - Beta(0.5, 0.5)	11.3	3.3	1.7	0.6	0.2	-
U3 - Beta(0.7, 0.3)	11.3	3.4	1.6	0.6	0.2	-
U3 - Beta(1, 1)	11.4	3.3	1.6	0.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(1, 1)	3.3	1.7	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						607.7
U2 - Beta(0.5, 0.5)						607.9
U2 - Beta(0.7, 0.3)						607.7
U2 - Beta(1, 1)						607.9
U3 - Beta(0.3, 0.7)						608
U3 - Beta(0.5, 0.5)						608.2
U3 - Beta(0.7, 0.3)						608
U3 - Beta(1, 1)						607.9
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	33.7	16.8	32.4	9.9	2.3	4.9
U2 - Beta(0.5, 0.5)	34.3	17.3	31.6	9.9	2.1	4.8
U2 - Beta(0.7, 0.3)	34.7	17.3	31.2	9.6	2.3	4.9
U2 - Beta(1, 1)	34.2	18.1	30.8	9.7	2.3	4.9
U3 - Beta(0.3, 0.7)	36.7	17.2	30.4	9.2	1.6	4.9
U3 - Beta(0.5, 0.5)	36.9	18.6	29.4	8.9	1.3	4.9
U3 - Beta(0.7, 0.3)	37.5	18.4	29	8.9	1.3	4.9
U3 - Beta(1, 1)	38.9	17.9	28.4	8.8	1.1	4.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	5.3	7.2	3.6	1.6	-
U2 - Beta(0.5, 0.5)	11.5	5.3	7.1	3.6	1.6	-
U2 - Beta(0.7, 0.3)	11.6	5.2	7.1	3.6	1.6	-
U2 - Beta(1, 1)	11.6	5.2	7.1	3.7	1.6	-
U3 - Beta(0.3, 0.7)	11.5	5.2	7.1	3.7	1.6	-
U3 - Beta(0.5, 0.5)	11.6	5.3	7.1	3.7	1.6	-
U3 - Beta(0.7, 0.3)	11.6	5.2	7.1	3.6	1.6	-
U3 - Beta(1, 1)	11.6	5.1	7.2	3.7	1.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	2.5	2	1	0.4	-
U2 - Beta(0.5, 0.5)	3.3	2.5	2	1	0.4	-
U2 - Beta(0.7, 0.3)	3.4	2.5	2	0.9	0.4	-
U2 - Beta(1, 1)	3.4	2.5	1.9	1	0.4	-
U3 - Beta(0.3, 0.7)	3.3	2.5	2	1	0.4	-
U3 - Beta(0.5, 0.5)	3.4	2.5	1.9	1	0.4	-
U3 - Beta(0.7, 0.3)	3.4	2.5	1.9	1	0.4	-
U3 - Beta(1, 1)	3.4	2.4	2	1	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1099.3
U2 - Beta(0.5, 0.5)						1098.9
U2 - Beta(0.7, 0.3)						1099
U2 - Beta(1, 1)						1098.6
U3 - Beta(0.3, 0.7)						1099.3
U3 - Beta(0.5, 0.5)						1099.3
U3 - Beta(0.7, 0.3)						1099.3
U3 - Beta(1, 1)						1099.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.1	41.6	18.2	8.2	1.4	18.5
U2 - Beta(0.5, 0.5)	12.7	41.7	18.3	6.9	1.9	18.5
U2 - Beta(0.7, 0.3)	12.6	41.4	19	6.7	1.8	18.5
U2 - Beta(1, 1)	13	41.3	19.4	6.6	1.4	18.3
U3 - Beta(0.3, 0.7)	13.5	43.3	17.1	6.3	1.3	18.5
U3 - Beta(0.5, 0.5)	13.7	43.3	17	6.1	1.3	18.6
U3 - Beta(0.7, 0.3)	14.8	43.6	16.9	5.1	1.2	18.4
U3 - Beta(1, 1)	14.6	43.7	16.4	5.4	1.2	18.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.5	7.6	6	3.4	1.5	-
U2 - Beta(0.5, 0.5)	8.6	7.7	5.9	3.3	1.5	-
U2 - Beta(0.7, 0.3)	8.6	7.8	5.8	3.4	1.5	-
U2 - Beta(1, 1)	8.5	7.7	5.9	3.3	1.5	-
U3 - Beta(0.3, 0.7)	8.5	7.6	6	3.4	1.6	-
U3 - Beta(0.5, 0.5)	8.6	7.7	5.8	3.4	1.5	-
U3 - Beta(0.7, 0.3)	8.5	7.8	6	3.4	1.5	-
U3 - Beta(1, 1)	8.6	7.7	5.9	3.3	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	3.3	1.7	1.1	0.5	-
U2 - Beta(0.5, 0.5)	2.3	3.3	1.7	1.1	0.4	-
U2 - Beta(0.7, 0.3)	2.3	3.3	1.6	1.1	0.4	-
U2 - Beta(1, 1)	2.3	3.3	1.7	1.1	0.4	-
U3 - Beta(0.3, 0.7)	2.3	3.3	1.7	1.1	0.4	-
U3 - Beta(0.5, 0.5)	2.3	3.3	1.6	1.1	0.4	-
U3 - Beta(0.7, 0.3)	2.3	3.3	1.7	1.1	0.4	-
U3 - Beta(1, 1)	2.3	3.3	1.6	1.1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1006
U2 - Beta(0.5, 0.5)						1004.8
U2 - Beta(0.7, 0.3)						1005.3
U2 - Beta(1, 1)						1005.7
U3 - Beta(0.3, 0.7)						1006.5
U3 - Beta(0.5, 0.5)						1005
U3 - Beta(0.7, 0.3)						1005.6
U3 - Beta(1, 1)						1004.6

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.9	7.6	22.2	44.3	15.5	3.5
U2 - Beta(0.5, 0.5)	7.2	7.6	23.4	42.8	15.4	3.6
U2 - Beta(0.7, 0.3)	8.7	8.6	21.5	42	15.6	3.6
U2 - Beta(1, 1)	7.9	8.6	21.9	42.8	15.2	3.6
U3 - Beta(0.3, 0.7)	8.7	9.8	22.1	42.2	13.6	3.6
U3 - Beta(0.5, 0.5)	11.1	9.2	21.4	41.2	13.5	3.6
U3 - Beta(0.7, 0.3)	12.7	10.1	19.3	41.6	12.6	3.7
U3 - Beta(1, 1)	13.5	8.8	19.6	41.5	12.8	3.8
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	4.2	6.7	8.1	5.9	-
U2 - Beta(0.5, 0.5)	4.4	4.3	6.8	8.1	5.8	-
U2 - Beta(0.7, 0.3)	4.4	4.3	6.7	8.2	5.7	-
U2 - Beta(1, 1)	4.4	4.3	6.6	8.1	5.9	-
U3 - Beta(0.3, 0.7)	4.4	4.2	6.8	8.2	5.8	-
U3 - Beta(0.5, 0.5)	4.4	4.3	6.7	8.1	5.8	-
U3 - Beta(0.7, 0.3)	4.5	4.3	6.7	8.2	5.7	-
U3 - Beta(1, 1)	4.4	4.3	6.7	8.2	5.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	2.7	2.1	2.7	1.4	-
U2 - Beta(0.5, 0.5)	0.6	2.8	2.1	2.7	1.4	-
U2 - Beta(0.7, 0.3)	0.7	2.8	2.1	2.7	1.4	-
U2 - Beta(1, 1)	0.7	2.8	2	2.7	1.4	-
U3 - Beta(0.3, 0.7)	0.6	2.7	2.1	2.7	1.4	-
U3 - Beta(0.5, 0.5)	0.7	2.8	2.1	2.7	1.4	-
U3 - Beta(0.7, 0.3)	0.7	2.8	2.1	2.7	1.4	-
U3 - Beta(1, 1)	0.7	2.8	2.1	2.7	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1090
U2 - Beta(0.5, 0.5)						1089.2
U2 - Beta(0.7, 0.3)						1088.9
U2 - Beta(1, 1)						1090.4
U3 - Beta(0.3, 0.7)						1089.9
U3 - Beta(0.5, 0.5)						1088.5
U3 - Beta(0.7, 0.3)						1087.8
U3 - Beta(1, 1)						1090.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.7	35	25.1	19.4	10.7	6.1
U2 - Beta(0.5, 0.5)	3.9	36.4	25.8	18.4	9.4	6.1
U2 - Beta(0.7, 0.3)	4	36.8	25.9	19	8.2	6.1
U2 - Beta(1, 1)	4.8	36.3	26.3	17.9	8.6	6.1
U3 - Beta(0.3, 0.7)	5	39.9	25.8	16.2	7	6.1
U3 - Beta(0.5, 0.5)	6.8	41.1	24.5	16.3	5.2	6.1
U3 - Beta(0.7, 0.3)	7.6	42.6	23.3	15.7	4.7	6.1
U3 - Beta(1, 1)	9.8	42.3	23.1	14.4	4.3	6.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	6.4	6.9	5.8	5.3	-
U2 - Beta(0.5, 0.5)	4.6	6.6	6.7	5.8	5.2	-
U2 - Beta(0.7, 0.3)	4.6	6.7	6.7	5.9	5.1	-
U2 - Beta(1, 1)	4.6	6.7	6.6	5.8	5.1	-
U3 - Beta(0.3, 0.7)	4.5	6.3	6.8	5.8	5.4	-
U3 - Beta(0.5, 0.5)	4.5	6.6	6.7	5.8	5.1	-
U3 - Beta(0.7, 0.3)	4.5	6.7	6.7	5.9	5.1	-
U3 - Beta(1, 1)	4.5	6.7	6.6	5.8	5.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	2.2	2.2	1.9	-
U2 - Beta(0.5, 0.5)	0.7	3.5	2.1	2.2	1.8	-
U2 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.8	-
U2 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.8	-
U3 - Beta(0.3, 0.7)	0.7	3.4	2.2	2.2	1.9	-
U3 - Beta(0.5, 0.5)	0.7	3.5	2.2	2.2	1.8	-
U3 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.8	-
U3 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016.3
U2 - Beta(0.5, 0.5)						1014.1
U2 - Beta(0.7, 0.3)						1013.9
U2 - Beta(1, 1)						1014.9
U3 - Beta(0.3, 0.7)						1016.4
U3 - Beta(0.5, 0.5)						1014.8
U3 - Beta(0.7, 0.3)						1014.3
U3 - Beta(1, 1)						1014.5
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	66.4	11.6	2.9	0.9	0.2	18
U2 - Beta(0.5, 0.5)	66.6	11.3	3	0.6	0.2	18.3
U2 - Beta(0.7, 0.3)	66	12.4	3.1	0.4	0.2	17.9
U2 - Beta(1, 1)	66.3	11.6	2.9	0.8	0.2	18.2
U3 - Beta(0.3, 0.7)	67.6	11.1	2.3	0.8	0.2	18
U3 - Beta(0.5, 0.5)	67.9	11.2	2.5	0.2	0.2	18
U3 - Beta(0.7, 0.3)	68.1	10.6	2.8	0.3	0.1	18.1
U3 - Beta(1, 1)	68.2	10.6	2.4	0.3	0.2	18.3
Mean administrations						
U2 - Beta(0.3, 0.7)	19.7	3.8	2.2	0.9	0.3	-
U2 - Beta(0.5, 0.5)	19.6	3.8	2.2	0.8	0.2	-
U2 - Beta(0.7, 0.3)	19.6	3.9	2.2	0.9	0.2	-
U2 - Beta(1, 1)	19.6	3.9	2.2	0.8	0.3	-
U3 - Beta(0.3, 0.7)	19.6	3.8	2.2	0.9	0.3	-
U3 - Beta(0.5, 0.5)	19.7	3.8	2.1	0.9	0.3	-
U3 - Beta(0.7, 0.3)	19.7	3.8	2.2	0.8	0.3	-
U3 - Beta(1, 1)	19.6	3.9	2.1	0.9	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.4	1.5	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	5.4	1.5	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	5.4	1.5	0.6	0.3	0.1	-
U2 - Beta(1, 1)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(1, 1)	5.4	1.5	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.1
U2 - Beta(0.5, 0.5)						1034.5
U2 - Beta(0.7, 0.3)						1037.5
U2 - Beta(1, 1)						1036.7
U3 - Beta(0.3, 0.7)						1037.9
U3 - Beta(0.5, 0.5)						1037
U3 - Beta(0.7, 0.3)						1037.6
U3 - Beta(1, 1)						1037

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.2	35.9	20.3	10.4	2.6	5.6
U2 - Beta(0.5, 0.5)	25.2	36.8	19.6	10.6	2.2	5.6
U2 - Beta(0.7, 0.3)	25.2	37.3	19.2	10.9	1.8	5.6
U2 - Beta(1, 1)	25.5	36.9	20.1	9.7	2.2	5.6
U3 - Beta(0.3, 0.7)	26.6	39.6	18	8.5	1.8	5.5
U3 - Beta(0.5, 0.5)	27.4	39.8	17	8.8	1.4	5.6
U3 - Beta(0.7, 0.3)	27.7	39.6	16.9	9.1	1.2	5.5
U3 - Beta(1, 1)	27.6	38.9	17.7	8.7	1.5	5.6
Mean administrations						
U2 - Beta(0.3, 0.7)	10.8	7	5.5	3.7	2	-
U2 - Beta(0.5, 0.5)	10.9	7	5.5	3.7	1.9	-
U2 - Beta(0.7, 0.3)	10.9	7.1	5.5	3.6	1.9	-
U2 - Beta(1, 1)	10.9	7	5.6	3.6	2	-
U3 - Beta(0.3, 0.7)	10.9	6.9	5.5	3.7	2	-
U3 - Beta(0.5, 0.5)	10.8	7.1	5.5	3.6	2	-
U3 - Beta(0.7, 0.3)	10.9	7.1	5.6	3.6	1.9	-
U3 - Beta(1, 1)	10.9	7	5.6	3.6	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.8	1.3	1.1	0.5	-
U2 - Beta(0.5, 0.5)	3	2.8	1.3	1	0.5	-
U2 - Beta(0.7, 0.3)	3	2.8	1.4	1	0.5	-
U2 - Beta(1, 1)	3	2.8	1.3	1	0.5	-
U3 - Beta(0.3, 0.7)	3.1	2.8	1.4	1	0.5	-
U3 - Beta(0.5, 0.5)	3	2.8	1.3	1	0.5	-
U3 - Beta(0.7, 0.3)	3.1	2.8	1.4	1	0.5	-
U3 - Beta(1, 1)	3	2.8	1.4	1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1114.8
U2 - Beta(0.5, 0.5)						1114.9
U2 - Beta(0.7, 0.3)						1113.7
U2 - Beta(1, 1)						1113.7
U3 - Beta(0.3, 0.7)						1114
U3 - Beta(0.5, 0.5)						1114.4
U3 - Beta(0.7, 0.3)						1114.1
U3 - Beta(1, 1)						1113.9
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.1	6.1	2	0.7	0.3	62.8
U2 - Beta(0.5, 0.5)	28.7	5.7	1.9	0.7	0.2	62.8
U2 - Beta(0.7, 0.3)	28.6	5.3	2.1	0.5	0.2	63.3
U2 - Beta(1, 1)	28.2	6.8	1.6	0.5	0.2	62.7
U3 - Beta(0.3, 0.7)	29.2	6.1	1.3	0.5	0.3	62.6
U3 - Beta(0.5, 0.5)	29.8	5.4	1.2	0.5	0.2	62.9
U3 - Beta(0.7, 0.3)	28.9	6.1	1.2	0.5	0.2	63.1
U3 - Beta(1, 1)	29	5.8	1.1	0.5	0.2	63.4
Mean administrations						
U2 - Beta(0.3, 0.7)	14	3.3	1.5	0.4	0.2	-
U2 - Beta(0.5, 0.5)	14	3.3	1.5	0.4	0.1	-
U2 - Beta(0.7, 0.3)	14	3.3	1.5	0.4	0.1	-
U2 - Beta(1, 1)	14.1	3.2	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	14	3.3	1.5	0.5	0.1	-
U3 - Beta(0.5, 0.5)	14	3.3	1.5	0.4	0.1	-
U3 - Beta(0.7, 0.3)	14	3.2	1.5	0.4	0.1	-
U3 - Beta(1, 1)	14.1	3.2	1.5	0.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	1.6	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	4.2	1.6	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	4.2	1.6	0.5	0.2	0	-
U2 - Beta(1, 1)	4.2	1.5	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	4.1	1.5	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	4.2	1.6	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	4.2	1.5	0.5	0.2	0	-
U3 - Beta(1, 1)	4.2	1.5	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						705.6
U2 - Beta(0.5, 0.5)						705.9
U2 - Beta(0.7, 0.3)						704.4
U2 - Beta(1, 1)						705.2
U3 - Beta(0.3, 0.7)						704.7
U3 - Beta(0.5, 0.5)						704.7
U3 - Beta(0.7, 0.3)						706.1
U3 - Beta(1, 1)						705.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.4	9.9	17	12.6	22.7	29.4
U2 - Beta(0.5, 0.5)	9.3	9.8	16.8	12.4	22.2	29.5
U2 - Beta(0.7, 0.3)	9.7	11.5	16.7	11.3	21.5	29.3
U2 - Beta(1, 1)	12.1	10.2	15.7	10.3	22.3	29.4
U3 - Beta(0.3, 0.7)	13.1	10.4	15.9	10	21.1	29.5
U3 - Beta(0.5, 0.5)	13.6	10.8	15.5	10	21	29.1
U3 - Beta(0.7, 0.3)	16.2	11.7	13.1	10.2	19.4	29.4
U3 - Beta(1, 1)	18.7	11.3	11.6	10.3	18.7	29.4
Mean administrations						
U2 - Beta(0.3, 0.7)	7.1	4.2	5.4	3.7	4.6	-
U2 - Beta(0.5, 0.5)	7.3	4.3	5.3	3.7	4.5	-
U2 - Beta(0.7, 0.3)	7.3	4.4	5.3	3.7	4.5	-
U2 - Beta(1, 1)	7.3	4.4	5.2	3.7	4.5	-
U3 - Beta(0.3, 0.7)	7.2	4.2	5.4	3.7	4.6	-
U3 - Beta(0.5, 0.5)	7.2	4.3	5.4	3.6	4.6	-
U3 - Beta(0.7, 0.3)	7.3	4.4	5.2	3.7	4.5	-
U3 - Beta(1, 1)	7.4	4.4	5.2	3.7	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	2.4	1.9	1.5	1.3	-
U2 - Beta(0.5, 0.5)	1.9	2.5	1.8	1.5	1.3	-
U2 - Beta(0.7, 0.3)	1.9	2.5	1.8	1.5	1.3	-
U2 - Beta(1, 1)	1.9	2.5	1.8	1.5	1.3	-
U3 - Beta(0.3, 0.7)	1.8	2.4	1.9	1.5	1.3	-
U3 - Beta(0.5, 0.5)	1.9	2.5	1.9	1.5	1.3	-
U3 - Beta(0.7, 0.3)	1.9	2.5	1.8	1.5	1.3	-
U3 - Beta(1, 1)	1.9	2.5	1.8	1.5	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						882.8
U2 - Beta(0.5, 0.5)						882.4
U2 - Beta(0.7, 0.3)						881.8
U2 - Beta(1, 1)						881.3
U3 - Beta(0.3, 0.7)						880.1
U3 - Beta(0.5, 0.5)						882.7
U3 - Beta(0.7, 0.3)						880.9
U3 - Beta(1, 1)						880.5

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.9	25.7	11	4	1.2	51.2
U2 - Beta(0.5, 0.5)	7.8	25.1	10.8	3.9	1.2	51.2
U2 - Beta(0.7, 0.3)	7.8	24.9	11.2	3.6	1	51.5
U2 - Beta(1, 1)	7.7	25.4	10.6	3.9	1.1	51.3
U3 - Beta(0.3, 0.7)	8.5	25.6	10.2	3.6	1	51.1
U3 - Beta(0.5, 0.5)	8.1	26.4	10	2.9	1.1	51.5
U3 - Beta(0.7, 0.3)	8.7	27.2	9.4	2.6	0.8	51.3
U3 - Beta(1, 1)	8.6	27.6	9	2.7	0.9	51.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	5.5	3.7	1.9	0.7	-
U2 - Beta(0.5, 0.5)	9.2	5.5	3.7	1.8	0.8	-
U2 - Beta(0.7, 0.3)	9.2	5.5	3.6	1.8	0.8	-
U2 - Beta(1, 1)	9.2	5.5	3.6	1.8	0.8	-
U3 - Beta(0.3, 0.7)	9.2	5.4	3.8	1.8	0.7	-
U3 - Beta(0.5, 0.5)	9.2	5.5	3.7	1.8	0.7	-
U3 - Beta(0.7, 0.3)	9.2	5.5	3.7	1.8	0.7	-
U3 - Beta(1, 1)	9.2	5.5	3.6	1.8	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	2.4	1	0.6	0.2	-
U2 - Beta(0.5, 0.5)	2.5	2.4	1	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.5	2.4	1	0.6	0.2	-
U2 - Beta(1, 1)	2.5	2.5	1	0.6	0.2	-
U3 - Beta(0.3, 0.7)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(0.5, 0.5)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(1, 1)	2.5	2.5	1	0.6	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						776.2
U2 - Beta(0.5, 0.5)						776.2
U2 - Beta(0.7, 0.3)						775.4
U2 - Beta(1, 1)						775.8
U3 - Beta(0.3, 0.7)						775.6
U3 - Beta(0.5, 0.5)						776.4
U3 - Beta(0.7, 0.3)						776.5
U3 - Beta(1, 1)						776.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	62.9	18.8	3.3	0.1	0	14.9
U2 - Beta(0.5, 0.5)	63.8	18.9	2.6	0.2	0	14.5
U2 - Beta(0.7, 0.3)	62.8	19.7	3	0.1	0	14.4
U2 - Beta(1, 1)	63.1	19.4	2.8	0.1	0	14.6
U3 - Beta(0.3, 0.7)	66.2	17.4	2.2	0	0	14.2
U3 - Beta(0.5, 0.5)	65.6	17.8	2	0.1	0	14.5
U3 - Beta(0.7, 0.3)	65.8	18.2	1.7	0.1	0	14.2
U3 - Beta(1, 1)	67	16.6	1.5	0.1	0	14.8
Mean administrations						
U2 - Beta(0.3, 0.7)	18.1	5.7	2.9	0.7	0.1	-
U2 - Beta(0.5, 0.5)	18.1	5.7	2.9	0.7	0.1	-
U2 - Beta(0.7, 0.3)	18.1	5.8	2.9	0.7	0.1	-
U2 - Beta(1, 1)	18.1	5.7	2.9	0.7	0.1	-
U3 - Beta(0.3, 0.7)	18	5.8	2.9	0.7	0.1	-
U3 - Beta(0.5, 0.5)	18.1	5.7	2.9	0.7	0.1	-
U3 - Beta(0.7, 0.3)	18.1	5.7	2.8	0.7	0.1	-
U3 - Beta(1, 1)	18.1	5.7	2.8	0.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(0.5, 0.5)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(0.7, 0.3)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(1, 1)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.3, 0.7)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.5, 0.5)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.7, 0.3)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(1, 1)	5.6	2.5	0.9	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						997.8
U2 - Beta(0.5, 0.5)						998.9
U2 - Beta(0.7, 0.3)						998.4
U2 - Beta(1, 1)						997.6
U3 - Beta(0.3, 0.7)						998.6
U3 - Beta(0.5, 0.5)						997.9
U3 - Beta(0.7, 0.3)						998.8
U3 - Beta(1, 1)						997
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.5	14	13.4	36.7	18	5.4
U2 - Beta(0.5, 0.5)	13.8	13.9	11.2	38.9	16.8	5.4
U2 - Beta(0.7, 0.3)	15.5	13.9	11.2	37.1	16.7	5.6
U2 - Beta(1, 1)	16.3	13.6	10.8	36.3	17.5	5.5
U3 - Beta(0.3, 0.7)	17	15.1	10.9	36.2	15.4	5.4
U3 - Beta(0.5, 0.5)	19.7	13.5	10.3	35.2	15.8	5.5
U3 - Beta(0.7, 0.3)	21.7	14.9	8.3	33.7	15.9	5.5
U3 - Beta(1, 1)	23.9	14.9	6.9	33.1	15.6	5.6
Mean administrations						
U2 - Beta(0.3, 0.7)	4.8	4.6	5.6	7.3	6.7	-
U2 - Beta(0.5, 0.5)	4.9	4.7	5.6	7.4	6.5	-
U2 - Beta(0.7, 0.3)	5	4.9	5.5	7.3	6.4	-
U2 - Beta(1, 1)	5	4.9	5.4	7.2	6.6	-
U3 - Beta(0.3, 0.7)	4.8	4.7	5.6	7.4	6.6	-
U3 - Beta(0.5, 0.5)	4.9	4.8	5.6	7.3	6.5	-
U3 - Beta(0.7, 0.3)	5	4.9	5.6	7.3	6.4	-
U3 - Beta(1, 1)	4.9	4.9	5.5	7.1	6.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	2.9	1.8	2.7	1.5	-
U2 - Beta(0.5, 0.5)	0.9	2.9	1.8	2.8	1.4	-
U2 - Beta(0.7, 0.3)	0.9	3	1.8	2.7	1.5	-
U2 - Beta(1, 1)	0.9	3	1.7	2.7	1.5	-
U3 - Beta(0.3, 0.7)	0.8	3	1.8	2.8	1.5	-
U3 - Beta(0.5, 0.5)	0.9	3	1.8	2.7	1.5	-
U3 - Beta(0.7, 0.3)	0.9	3	1.8	2.7	1.5	-
U3 - Beta(1, 1)	0.9	3	1.7	2.7	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.3
U2 - Beta(0.5, 0.5)						1059.8
U2 - Beta(0.7, 0.3)						1057.7
U2 - Beta(1, 1)						1059.3
U3 - Beta(0.3, 0.7)						1062.2
U3 - Beta(0.5, 0.5)						1058.9
U3 - Beta(0.7, 0.3)						1058.1
U3 - Beta(1, 1)						1060.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.3	27.6	11.6	4	1.3	46.2
U2 - Beta(0.5, 0.5)	9.3	28.1	11.1	4	1.2	46.3
U2 - Beta(0.7, 0.3)	9.7	27.1	11.7	4.2	1.2	46.1
U2 - Beta(1, 1)	9.9	27.8	11.4	3.7	1.1	46.1
U3 - Beta(0.3, 0.7)	10.1	28.8	10.3	3.6	0.9	46.3
U3 - Beta(0.5, 0.5)	11	28.8	9.9	3.1	1.1	46.1
U3 - Beta(0.7, 0.3)	11	28.3	10.3	3.3	0.9	46.2
U3 - Beta(1, 1)	11.7	28.6	9.8	2.9	0.9	46.1
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	6	4	2	0.8	-
U2 - Beta(0.5, 0.5)	9	6	4	1.9	0.8	-
U2 - Beta(0.7, 0.3)	9	6	3.9	2	0.8	-
U2 - Beta(1, 1)	9	6	3.9	2	0.8	-
U3 - Beta(0.3, 0.7)	8.9	5.9	4	2	0.8	-
U3 - Beta(0.5, 0.5)	9	5.9	4	2	0.8	-
U3 - Beta(0.7, 0.3)	9	6	3.9	2	0.8	-
U3 - Beta(1, 1)	9	6	3.9	2	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	2.8	1.2	0.7	0.2	-
U2 - Beta(0.5, 0.5)	2.4	2.8	1.2	0.7	0.2	-
U2 - Beta(0.7, 0.3)	2.5	2.8	1.2	0.7	0.2	-
U2 - Beta(1, 1)	2.4	2.8	1.2	0.7	0.2	-
U3 - Beta(0.3, 0.7)	2.4	2.7	1.2	0.7	0.2	-
U3 - Beta(0.5, 0.5)	2.5	2.7	1.2	0.7	0.2	-
U3 - Beta(0.7, 0.3)	2.5	2.8	1.2	0.7	0.2	-
U3 - Beta(1, 1)	2.5	2.8	1.2	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						789.9
U2 - Beta(0.5, 0.5)						788
U2 - Beta(0.7, 0.3)						789
U2 - Beta(1, 1)						788.9
U3 - Beta(0.3, 0.7)						789.4
U3 - Beta(0.5, 0.5)						788.7
U3 - Beta(0.7, 0.3)						788
U3 - Beta(1, 1)						789.4

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.1	14.8	16.2	26.8	13.9	14.2
U2 - Beta(0.5, 0.5)	18.7	13.6	15.4	24.5	13.2	14.6
U2 - Beta(0.7, 0.3)	19.2	14.7	13.9	25.9	12.2	14.1
U2 - Beta(1, 1)	20.2	15.7	12.6	24.2	12.6	14.7
U3 - Beta(0.3, 0.7)	22.3	15.3	12.5	24.8	10.8	14.3
U3 - Beta(0.5, 0.5)	25.2	15.6	11.1	23.9	9.7	14.5
U3 - Beta(0.7, 0.3)	26.8	17.7	10.1	22.5	8.8	14.1
U3 - Beta(1, 1)	31.3	16.4	8.2	21.6	8	14.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.2	5.2	5.9	5.7	4.7	-
U2 - Beta(0.5, 0.5)	6.3	5.3	5.9	5.7	4.6	-
U2 - Beta(0.7, 0.3)	6.3	5.5	5.8	5.7	4.5	-
U2 - Beta(1, 1)	6.4	5.5	5.7	5.6	4.6	-
U3 - Beta(0.3, 0.7)	6.1	5.2	6	5.7	4.7	-
U3 - Beta(0.5, 0.5)	6.2	5.3	6	5.7	4.6	-
U3 - Beta(0.7, 0.3)	6.4	5.5	5.7	5.7	4.5	-
U3 - Beta(1, 1)	6.3	5.5	5.7	5.7	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.1	2.1	2.3	1.4	-
U2 - Beta(0.5, 0.5)	1.5	3.1	2	2.2	1.4	-
U2 - Beta(0.7, 0.3)	1.5	3.2	2	2.2	1.4	-
U2 - Beta(1, 1)	1.5	3.2	2	2.2	1.4	-
U3 - Beta(0.3, 0.7)	1.4	3.1	2.1	2.2	1.4	-
U3 - Beta(0.5, 0.5)	1.4	3.1	2.1	2.2	1.4	-
U3 - Beta(0.7, 0.3)	1.5	3.2	2	2.2	1.3	-
U3 - Beta(1, 1)	1.5	3.2	2	2.2	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						966.6
U2 - Beta(0.5, 0.5)						967.2
U2 - Beta(0.7, 0.3)						967
U2 - Beta(1, 1)						965.4
U3 - Beta(0.3, 0.7)						967
U3 - Beta(0.5, 0.5)						966.2
U3 - Beta(0.7, 0.3)						966.1
U3 - Beta(1, 1)						964.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.1	26.7	10.4	2.8	0.5	19.5
U2 - Beta(0.5, 0.5)	42.8	24.7	9.6	3.1	0.2	19.6
U2 - Beta(0.7, 0.3)	43.5	24.3	9.7	2.6	0.3	19.6
U2 - Beta(1, 1)	45.3	23.8	8.6	2.4	0.1	19.8
U3 - Beta(0.3, 0.7)	46.6	24.9	7.1	1.8	0.1	19.5
U3 - Beta(0.5, 0.5)	50.2	21.9	6.5	1.5	0.3	19.6
U3 - Beta(0.7, 0.3)	52.2	21.5	5.3	1	0.3	19.7
U3 - Beta(1, 1)	53.7	21.3	4.4	1	0	19.6
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	7	5.8	3	1.2	-
U2 - Beta(0.5, 0.5)	10.6	7.1	5.7	2.9	1.1	-
U2 - Beta(0.7, 0.3)	10.6	7.1	5.7	2.9	1.2	-
U2 - Beta(1, 1)	10.7	7.1	5.6	2.8	1.2	-
U3 - Beta(0.3, 0.7)	10.5	6.9	5.8	3	1.2	-
U3 - Beta(0.5, 0.5)	10.5	7	5.7	3	1.2	-
U3 - Beta(0.7, 0.3)	10.7	7	5.6	2.9	1.2	-
U3 - Beta(1, 1)	10.7	7.2	5.6	2.9	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.5	2	1.3	0.5	-
U2 - Beta(0.5, 0.5)	3.2	3.6	2	1.3	0.5	-
U2 - Beta(0.7, 0.3)	3.2	3.6	2	1.3	0.5	-
U2 - Beta(1, 1)	3.3	3.6	2	1.2	0.5	-
U3 - Beta(0.3, 0.7)	3.2	3.5	2	1.3	0.5	-
U3 - Beta(0.5, 0.5)	3.2	3.5	2	1.3	0.5	-
U3 - Beta(0.7, 0.3)	3.3	3.5	2	1.3	0.5	-
U3 - Beta(1, 1)	3.3	3.6	1.9	1.3	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						929.1
U2 - Beta(0.5, 0.5)						929.1
U2 - Beta(0.7, 0.3)						929.3
U2 - Beta(1, 1)						929.1
U3 - Beta(0.3, 0.7)						929.2
U3 - Beta(0.5, 0.5)						928.9
U3 - Beta(0.7, 0.3)						929.2
U3 - Beta(1, 1)						929.9
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.2	25.3	17.5	17.1	12.8	0.1
U2 - Beta(0.5, 0.5)	29	26	16.2	15.7	13	0.1
U2 - Beta(0.7, 0.3)	29.9	27.6	16.1	15.3	11	0.1
U2 - Beta(1, 1)	30.7	27.2	15.9	15	11.1	0.1
U3 - Beta(0.3, 0.7)	34.8	30.1	15.3	13.6	6.1	0.1
U3 - Beta(0.5, 0.5)	38.9	28.3	13.5	13	6.2	0.1
U3 - Beta(0.7, 0.3)	38.3	28.8	14.9	13.3	4.6	0.1
U3 - Beta(1, 1)	42.6	28.8	13.3	10.8	4.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	5.5	5.5	5.8	7.4	-
U2 - Beta(0.5, 0.5)	5.8	5.7	5.5	5.8	7.2	-
U2 - Beta(0.7, 0.3)	5.8	5.8	5.5	5.9	7	-
U2 - Beta(1, 1)	5.8	5.8	5.5	5.8	7	-
U3 - Beta(0.3, 0.7)	5.7	5.5	5.5	5.8	7.5	-
U3 - Beta(0.5, 0.5)	5.8	5.7	5.5	5.8	7.1	-
U3 - Beta(0.7, 0.3)	5.8	5.7	5.5	5.9	7	-
U3 - Beta(1, 1)	5.8	5.8	5.5	5.9	7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.1	1.9	2.7	2.8	-
U2 - Beta(0.5, 0.5)	1.2	3.2	1.9	2.6	2.7	-
U2 - Beta(0.7, 0.3)	1.2	3.2	1.9	2.6	2.7	-
U2 - Beta(1, 1)	1.2	3.2	1.9	2.6	2.7	-
U3 - Beta(0.3, 0.7)	1.2	3.1	1.9	2.6	2.8	-
U3 - Beta(0.5, 0.5)	1.2	3.2	1.9	2.7	2.7	-
U3 - Beta(0.7, 0.3)	1.2	3.2	1.9	2.7	2.7	-
U3 - Beta(1, 1)	1.2	3.2	1.9	2.7	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008.5
U2 - Beta(0.5, 0.5)						1007.4
U2 - Beta(0.7, 0.3)						1007.7
U2 - Beta(1, 1)						1007.3
U3 - Beta(0.3, 0.7)						1008.9
U3 - Beta(0.5, 0.5)						1008
U3 - Beta(0.7, 0.3)						1007.9
U3 - Beta(1, 1)						1007.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.8	49.6	16.7	1.1	0	0.8
U2 - Beta(0.5, 0.5)	31.2	51.8	14.6	1.5	0	0.9
U2 - Beta(0.7, 0.3)	31.1	50.7	16	1.3	0	0.9
U2 - Beta(1, 1)	30.9	50.6	16.2	1.4	0	0.9
U3 - Beta(0.3, 0.7)	33.5	51.4	13.3	1	0	0.8
U3 - Beta(0.5, 0.5)	33.3	51.9	12.8	1.1	0	0.9
U3 - Beta(0.7, 0.3)	34.2	51	12.7	1.2	0	0.9
U3 - Beta(1, 1)	34.4	51.2	12.4	1.2	0	0.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.6	10	7.1	1.9	0.3	-
U2 - Beta(0.5, 0.5)	10.6	10.1	6.8	2	0.3	-
U2 - Beta(0.7, 0.3)	10.7	10.2	6.8	1.9	0.3	-
U2 - Beta(1, 1)	10.6	10.1	6.8	2	0.3	-
U3 - Beta(0.3, 0.7)	10.6	10.1	6.9	2	0.3	-
U3 - Beta(0.5, 0.5)	10.6	10.1	6.9	2	0.3	-
U3 - Beta(0.7, 0.3)	10.7	10.2	6.7	1.9	0.3	-
U3 - Beta(1, 1)	10.6	10.1	6.9	1.9	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	3.4	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.7	3.5	1.5	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.7	3.5	1.5	0.5	0.1	-
U2 - Beta(1, 1)	2.7	3.5	1.5	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.7	3.4	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.7	3.5	1.5	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.7	3.5	1.5	0.6	0.1	-
U3 - Beta(1, 1)	2.7	3.5	1.5	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1188
U2 - Beta(0.5, 0.5)						1187.6
U2 - Beta(0.7, 0.3)						1187.2
U2 - Beta(1, 1)						1186.3
U3 - Beta(0.3, 0.7)						1187.5
U3 - Beta(0.5, 0.5)						1187.2
U3 - Beta(0.7, 0.3)						1186.5
U3 - Beta(1, 1)						1186.4
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	10.6	11.3	13.9	55.3	1
U2 - Beta(0.5, 0.5)	9.1	9.2	10.8	14	55.8	1.1
U2 - Beta(0.7, 0.3)	10	10.5	10.3	14.1	54	1.1
U2 - Beta(1, 1)	10.3	10.7	10.4	13	54.5	1.1
U3 - Beta(0.3, 0.7)	14.1	12	9.9	12.7	50.3	1
U3 - Beta(0.5, 0.5)	16.1	12.7	9.4	12.6	48.1	1.1
U3 - Beta(0.7, 0.3)	17.4	13.5	9.9	12.2	46	1
U3 - Beta(1, 1)	18.7	14	9.7	11.2	45.4	1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	3.8	4.6	5.5	12.3	-
U2 - Beta(0.5, 0.5)	3.7	3.9	4.7	5.6	12	-
U2 - Beta(0.7, 0.3)	3.7	4	4.7	5.6	11.8	-
U2 - Beta(1, 1)	3.7	3.9	4.7	5.5	12	-
U3 - Beta(0.3, 0.7)	3.7	3.7	4.6	5.6	12.3	-
U3 - Beta(0.5, 0.5)	3.7	3.9	4.7	5.6	12	-
U3 - Beta(0.7, 0.3)	3.7	4	4.8	5.5	11.9	-
U3 - Beta(1, 1)	3.7	4	4.7	5.5	12	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	1.5	2.5	3.4	-
U2 - Beta(0.5, 0.5)	0.3	2.7	1.5	2.5	3.3	-
U2 - Beta(0.7, 0.3)	0.3	2.7	1.5	2.5	3.3	-
U2 - Beta(1, 1)	0.3	2.7	1.6	2.5	3.3	-
U3 - Beta(0.3, 0.7)	0.3	2.6	1.5	2.5	3.4	-
U3 - Beta(0.5, 0.5)	0.3	2.7	1.6	2.5	3.3	-
U3 - Beta(0.7, 0.3)	0.3	2.7	1.6	2.5	3.3	-
U3 - Beta(1, 1)	0.3	2.7	1.5	2.5	3.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1074.7
U2 - Beta(0.5, 0.5)						1070.3
U2 - Beta(0.7, 0.3)						1068
U2 - Beta(1, 1)						1069.3
U3 - Beta(0.3, 0.7)						1074
U3 - Beta(0.5, 0.5)						1070.3
U3 - Beta(0.7, 0.3)						1070.1
U3 - Beta(1, 1)						1070.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.9	49.9	14.1	1	0	17.1
U2 - Beta(0.5, 0.5)	18.4	49.9	13.7	0.9	0	17.1
U2 - Beta(0.7, 0.3)	18.9	48.7	14.1	1.1	0	17.2
U2 - Beta(1, 1)	19.2	50.9	11.7	1.1	0	17.1
U3 - Beta(0.3, 0.7)	20	50.7	11.6	0.4	0	17.3
U3 - Beta(0.5, 0.5)	20.3	50.7	11.2	0.6	0	17.2
U3 - Beta(0.7, 0.3)	21.1	49.4	11.4	0.7	0	17.4
U3 - Beta(1, 1)	22.5	49.1	10.4	0.7	0	17.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	10.4	6.6	1.8	0.2	-
U2 - Beta(0.5, 0.5)	8.2	10.5	6.5	1.7	0.3	-
U2 - Beta(0.7, 0.3)	8.2	10.6	6.3	1.7	0.3	-
U2 - Beta(1, 1)	8.2	10.4	6.4	1.7	0.3	-
U3 - Beta(0.3, 0.7)	8.1	10.4	6.6	1.7	0.3	-
U3 - Beta(0.5, 0.5)	8.1	10.6	6.4	1.7	0.3	-
U3 - Beta(0.7, 0.3)	8.2	10.7	6.2	1.7	0.3	-
U3 - Beta(1, 1)	8.2	10.6	6.4	1.7	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	4.5	1.9	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.1	4.5	1.8	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.1	4.6	1.8	0.6	0.1	-
U2 - Beta(1, 1)	2.1	4.5	1.8	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.1	4.5	1.9	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.1	4.6	1.8	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.1	4.6	1.8	0.6	0.1	-
U3 - Beta(1, 1)	2.1	4.6	1.8	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						987.3
U2 - Beta(0.5, 0.5)						986.3
U2 - Beta(0.7, 0.3)						986.4
U2 - Beta(1, 1)						986.8
U3 - Beta(0.3, 0.7)						986.7
U3 - Beta(0.5, 0.5)						986.1
U3 - Beta(0.7, 0.3)						986.2
U3 - Beta(1, 1)						986
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	45.7	28.1	8.3	0.8	7
U2 - Beta(0.5, 0.5)	9.8	45.8	28.8	8.1	0.7	6.8
U2 - Beta(0.7, 0.3)	11.6	46.5	25.4	9	0.9	6.6
U2 - Beta(1, 1)	11.2	46.9	25.2	9	0.9	6.8
U3 - Beta(0.3, 0.7)	13.5	49.8	23.2	6.3	0.4	6.8
U3 - Beta(0.5, 0.5)	15.5	48.2	21.6	7.2	0.7	6.8
U3 - Beta(0.7, 0.3)	17.7	47.3	21.1	6.5	0.6	6.8
U3 - Beta(1, 1)	20.7	47.7	18.8	5.4	0.4	7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	8.8	8.6	4.1	1.8	-
U2 - Beta(0.5, 0.5)	5.3	9	8.5	4.1	1.8	-
U2 - Beta(0.7, 0.3)	5.3	9.2	8.3	4.1	1.8	-
U2 - Beta(1, 1)	5.4	9.1	8.4	4	1.7	-
U3 - Beta(0.3, 0.7)	5.3	8.8	8.6	4.1	1.8	-
U3 - Beta(0.5, 0.5)	5.3	9.1	8.3	4.1	1.7	-
U3 - Beta(0.7, 0.3)	5.3	9.2	8.4	4.1	1.7	-
U3 - Beta(1, 1)	5.4	9.2	8.3	4.1	1.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	4.3	2.7	1.6	0.6	-
U2 - Beta(0.5, 0.5)	1	4.4	2.6	1.5	0.6	-
U2 - Beta(0.7, 0.3)	1	4.4	2.6	1.6	0.6	-
U2 - Beta(1, 1)	1.1	4.5	2.6	1.5	0.6	-
U3 - Beta(0.3, 0.7)	1	4.3	2.7	1.5	0.6	-
U3 - Beta(0.5, 0.5)	1.1	4.4	2.6	1.6	0.6	-
U3 - Beta(0.7, 0.3)	1	4.4	2.6	1.6	0.6	-
U3 - Beta(1, 1)	1.1	4.5	2.6	1.5	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1017.5
U2 - Beta(0.5, 0.5)						1015.6
U2 - Beta(0.7, 0.3)						1014.7
U2 - Beta(1, 1)						1015.4
U3 - Beta(0.3, 0.7)						1017.2
U3 - Beta(0.5, 0.5)						1014.8
U3 - Beta(0.7, 0.3)						1014.8
U3 - Beta(1, 1)						1014.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	54.8	0.7	0	0	0	44.5
U2 - Beta(0.5, 0.5)	54.9	0.7	0	0	0	44.4
U2 - Beta(0.7, 0.3)	54.5	0.8	0	0	0	44.7
U2 - Beta(1, 1)	54.6	0.8	0	0	0	44.6
U3 - Beta(0.3, 0.7)	54.8	0.6	0	0	0	44.6
U3 - Beta(0.5, 0.5)	54.5	0.8	0	0	0	44.7
U3 - Beta(0.7, 0.3)	54.2	1	0	0	0	44.8
U3 - Beta(1, 1)	54	0.8	0	0	0	45.2
Mean administrations						
U2 - Beta(0.3, 0.7)	20.4	1.8	0.4	0	0	-
U2 - Beta(0.5, 0.5)	20.4	1.9	0.4	0	0	-
U2 - Beta(0.7, 0.3)	20.4	1.9	0.4	0	0	-
U2 - Beta(1, 1)	20.4	1.9	0.4	0	0	-
U3 - Beta(0.3, 0.7)	20.4	1.8	0.5	0	0	-
U3 - Beta(0.5, 0.5)	20.4	1.9	0.4	0	0	-
U3 - Beta(0.7, 0.3)	20.4	1.9	0.4	0	0	-
U3 - Beta(1, 1)	20.3	1.8	0.4	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	0.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.2	0.6	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4.1	0.6	0.1	0	0	-
U2 - Beta(1, 1)	4.1	0.6	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.2	0.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.1	0.6	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4.1	0.6	0.1	0	0	-
U3 - Beta(1, 1)	4.1	0.6	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						987.4
U2 - Beta(0.5, 0.5)						987.5
U2 - Beta(0.7, 0.3)						988.5
U2 - Beta(1, 1)						988.8
U3 - Beta(0.3, 0.7)						989.3
U3 - Beta(0.5, 0.5)						988
U3 - Beta(0.7, 0.3)						988.7
U3 - Beta(1, 1)						986

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	24.1	54.5	9.8	1.2	0	10.4
U2 - Beta(0.5, 0.5)	23.3	54.3	10.5	1.3	0	10.6
U2 - Beta(0.7, 0.3)	24	52.8	10.8	1.8	0	10.6
U2 - Beta(1, 1)	24.7	52.9	11	1.3	0	10.1
U3 - Beta(0.3, 0.7)	27	53.5	7.8	1	0	10.7
U3 - Beta(0.5, 0.5)	27.9	52.5	8.2	1.1	0	10.3
U3 - Beta(0.7, 0.3)	28.6	51.2	8.4	1.3	0	10.5
U3 - Beta(1, 1)	29.5	50.1	8.7	1.2	0	10.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8	12.6	5.9	1.3	0.2	-
U2 - Beta(0.5, 0.5)	8.1	12.6	5.9	1.3	0.2	-
U2 - Beta(0.7, 0.3)	8.2	12.5	5.8	1.3	0.2	-
U2 - Beta(1, 1)	8.2	12.6	5.8	1.2	0.2	-
U3 - Beta(0.3, 0.7)	7.9	12.7	5.9	1.3	0.2	-
U3 - Beta(0.5, 0.5)	8	12.7	5.9	1.3	0.2	-
U3 - Beta(0.7, 0.3)	8.2	12.6	5.8	1.3	0.2	-
U3 - Beta(1, 1)	8.3	12.5	5.9	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	5.4	1.8	0.5	0	-
U2 - Beta(0.5, 0.5)	2	5.4	1.8	0.5	0	-
U2 - Beta(0.7, 0.3)	2.1	5.4	1.8	0.5	0	-
U2 - Beta(1, 1)	2.1	5.5	1.7	0.4	0	-
U3 - Beta(0.3, 0.7)	2	5.5	1.7	0.5	0	-
U3 - Beta(0.5, 0.5)	2	5.5	1.7	0.5	0	-
U3 - Beta(0.7, 0.3)	2	5.5	1.7	0.5	0	-
U3 - Beta(1, 1)	2.1	5.4	1.7	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1005.9
U2 - Beta(0.5, 0.5)						1005.7
U2 - Beta(0.7, 0.3)						1004.5
U2 - Beta(1, 1)						1005.9
U3 - Beta(0.3, 0.7)						1008
U3 - Beta(0.5, 0.5)						1007.6
U3 - Beta(0.7, 0.3)						1004.9
U3 - Beta(1, 1)						1006.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.3	3	0	0	0	90.7
U2 - Beta(0.5, 0.5)	6.3	3.1	0	0	0	90.6
U2 - Beta(0.7, 0.3)	6.4	3.2	0	0	0	90.4
U2 - Beta(1, 1)	6.6	3.4	0	0	0	90
U3 - Beta(0.3, 0.7)	6.4	3.2	0	0	0	90.4
U3 - Beta(0.5, 0.5)	6.3	3.1	0	0	0	90.6
U3 - Beta(0.7, 0.3)	6.5	3.3	0	0	0	90.2
U3 - Beta(1, 1)	6.7	3.3	0	0	0	90
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.1	0.4	0	0	-
U2 - Beta(0.5, 0.5)	11.7	2.1	0.4	0	0	-
U2 - Beta(0.7, 0.3)	11.7	2.1	0.4	0	0	-
U2 - Beta(1, 1)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.3, 0.7)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.5, 0.5)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.7, 0.3)	11.7	2.1	0.4	0	0	-
U3 - Beta(1, 1)	11.7	2.1	0.4	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	1.1	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.4	1.1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1.1	0.1	0	0	-
U2 - Beta(1, 1)	3.4	1.1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.4	1.1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.3	1.1	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.4	1.1	0.1	0	0	-
U3 - Beta(1, 1)	3.4	1.1	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						518.8
U2 - Beta(0.5, 0.5)						517.6
U2 - Beta(0.7, 0.3)						517.8
U2 - Beta(1, 1)						517.9
U3 - Beta(0.3, 0.7)						518.6
U3 - Beta(0.5, 0.5)						517.5
U3 - Beta(0.7, 0.3)						517.9
U3 - Beta(1, 1)						518.5
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.4	1.1	4.2	45	49.2	0.1
U2 - Beta(0.5, 0.5)	0.5	1.7	4.4	46.4	46.9	0.1
U2 - Beta(0.7, 0.3)	1.6	1.7	5	45.7	45.9	0.1
U2 - Beta(1, 1)	1.5	2	4.7	45.8	45.9	0.1
U3 - Beta(0.3, 0.7)	1.5	2.3	5.6	47.1	43.4	0.1
U3 - Beta(0.5, 0.5)	2.1	2.6	6	48.2	41	0.1
U3 - Beta(0.7, 0.3)	2.9	2.8	5.6	47.7	40.8	0.2
U3 - Beta(1, 1)	3.1	2.9	6.8	48.7	38.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.1	3.7	8.1	11.9	-
U2 - Beta(0.5, 0.5)	3.2	3.2	3.7	8.3	11.5	-
U2 - Beta(0.7, 0.3)	3.2	3.2	3.7	8.5	11.3	-
U2 - Beta(1, 1)	3.2	3.2	3.7	8.4	11.5	-
U3 - Beta(0.3, 0.7)	3.2	3.1	3.7	8.2	11.8	-
U3 - Beta(0.5, 0.5)	3.2	3.2	3.7	8.4	11.5	-
U3 - Beta(0.7, 0.3)	3.2	3.2	3.8	8.5	11.3	-
U3 - Beta(1, 1)	3.2	3.2	3.7	8.4	11.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.2	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.2	3.2	3.2	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.3	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1100.7
U2 - Beta(0.5, 0.5)						1097.9
U2 - Beta(0.7, 0.3)						1098.3
U2 - Beta(1, 1)						1098.7
U3 - Beta(0.3, 0.7)						1100.4
U3 - Beta(0.5, 0.5)						1098.5
U3 - Beta(0.7, 0.3)						1097.5
U3 - Beta(1, 1)						1098.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34.3	32.4	27.8	3.6	0	1.9
U2 - Beta(0.5, 0.5)	33.3	32.2	28.4	4.1	0	2
U2 - Beta(0.7, 0.3)	33	31.5	29.2	4.4	0	1.9
U2 - Beta(1, 1)	33.8	31.6	28.2	4.3	0	2.1
U3 - Beta(0.3, 0.7)	39.7	31.4	24.2	2.7	0	2
U3 - Beta(0.5, 0.5)	39.7	31.5	23.8	3.1	0	1.9
U3 - Beta(0.7, 0.3)	39.4	30.7	24.3	3.6	0	2
U3 - Beta(1, 1)	39.6	30.6	24.5	3.2	0	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	8.3	9.3	2.4	0.4	-
U2 - Beta(0.5, 0.5)	9.4	8.5	9	2.3	0.4	-
U2 - Beta(0.7, 0.3)	9.5	8.5	8.9	2.3	0.4	-
U2 - Beta(1, 1)	9.5	8.4	8.9	2.4	0.4	-
U3 - Beta(0.3, 0.7)	9.3	8.3	9.3	2.4	0.4	-
U3 - Beta(0.5, 0.5)	9.5	8.5	9	2.3	0.4	-
U3 - Beta(0.7, 0.3)	9.5	8.5	8.9	2.3	0.4	-
U3 - Beta(1, 1)	9.5	8.4	9	2.4	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.6	2.5	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.6	3.7	2.5	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.6	3.7	2.5	0.8	0.1	-
U2 - Beta(1, 1)	2.6	3.7	2.5	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.7	2.5	0.8	0.1	-
U3 - Beta(0.5, 0.5)	2.6	3.7	2.5	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.6	3.7	2.4	0.8	0.1	-
U3 - Beta(1, 1)	2.6	3.7	2.5	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1107.6
U2 - Beta(0.5, 0.5)						1105.9
U2 - Beta(0.7, 0.3)						1104.6
U2 - Beta(1, 1)						1104.9
U3 - Beta(0.3, 0.7)						1107.9
U3 - Beta(0.5, 0.5)						1105.5
U3 - Beta(0.7, 0.3)						1105.1
U3 - Beta(1, 1)						1105.5
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.8	28.3	25.7	3.2	0	20
U2 - Beta(0.5, 0.5)	22.2	29.6	25.2	3.1	0	19.9
U2 - Beta(0.7, 0.3)	23.2	28.7	23.9	4.1	0	20.1
U2 - Beta(1, 1)	22.6	28.8	24	4.5	0	20.1
U3 - Beta(0.3, 0.7)	25.7	28.1	23.3	2.7	0	20.2
U3 - Beta(0.5, 0.5)	26	28	21.8	3.8	0	20.4
U3 - Beta(0.7, 0.3)	27.4	27.9	20.7	3.7	0	20.3
U3 - Beta(1, 1)	27.9	27.8	20.6	3.5	0	20.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	6.9	7.7	2.3	0.3	-
U2 - Beta(0.5, 0.5)	9.9	6.9	7.5	2.3	0.3	-
U2 - Beta(0.7, 0.3)	9.9	7	7.4	2.3	0.3	-
U2 - Beta(1, 1)	9.9	6.8	7.5	2.3	0.3	-
U3 - Beta(0.3, 0.7)	9.7	6.7	7.8	2.4	0.3	-
U3 - Beta(0.5, 0.5)	9.9	6.8	7.6	2.3	0.3	-
U3 - Beta(0.7, 0.3)	10	6.9	7.4	2.3	0.3	-
U3 - Beta(1, 1)	9.9	6.8	7.5	2.3	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.3	2.3	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.9	3.3	2.2	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.9	3.3	2.2	0.8	0.1	-
U2 - Beta(1, 1)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.8	3.3	2.3	0.8	0.1	-
U3 - Beta(0.5, 0.5)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(1, 1)	2.9	3.3	2.2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						973.9
U2 - Beta(0.5, 0.5)						973.6
U2 - Beta(0.7, 0.3)						973.2
U2 - Beta(1, 1)						973.9
U3 - Beta(0.3, 0.7)						975.9
U3 - Beta(0.5, 0.5)						971.7
U3 - Beta(0.7, 0.3)						971.8
U3 - Beta(1, 1)						974.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.6	24.6	39.9	6.5	0.3	12.1
U2 - Beta(0.5, 0.5)	18	25.1	37.4	6.9	0.3	12.3
U2 - Beta(0.7, 0.3)	17.1	25	38.9	6.1	0.3	12.6
U2 - Beta(1, 1)	18.4	24.9	38.1	6.1	0.2	12.3
U3 - Beta(0.3, 0.7)	20.8	25.1	36.8	5	0.1	12.2
U3 - Beta(0.5, 0.5)	20.8	25.2	36.4	5.4	0.1	12.1
U3 - Beta(0.7, 0.3)	21.5	26.2	34.3	5.5	0.3	12.2
U3 - Beta(1, 1)	23.2	25.8	33.5	5.1	0.2	12.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	6.7	9.1	3.6	1.1	-
U2 - Beta(0.5, 0.5)	7.6	6.7	9.1	3.6	1.1	-
U2 - Beta(0.7, 0.3)	7.7	6.8	9	3.5	1.1	-
U2 - Beta(1, 1)	7.7	6.7	9	3.6	1.1	-
U3 - Beta(0.3, 0.7)	7.6	6.6	9.1	3.6	1.1	-
U3 - Beta(0.5, 0.5)	7.6	6.7	9.1	3.6	1.1	-
U3 - Beta(0.7, 0.3)	7.6	6.8	9.1	3.5	1	-
U3 - Beta(1, 1)	7.7	6.6	9.2	3.5	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(0.5, 0.5)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(0.7, 0.3)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(1, 1)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.3, 0.7)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.5, 0.5)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.7, 0.3)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(1, 1)	2	3.3	2.6	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.3
U2 - Beta(0.5, 0.5)						1037.9
U2 - Beta(0.7, 0.3)						1036.6
U2 - Beta(1, 1)						1036.7
U3 - Beta(0.3, 0.7)						1038.1
U3 - Beta(0.5, 0.5)						1037.9
U3 - Beta(0.7, 0.3)						1036.9
U3 - Beta(1, 1)						1036.4
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.8	31.3	4.3	0	0	47.6
U2 - Beta(0.5, 0.5)	16.7	31.1	4.8	0.1	0	47.3
U2 - Beta(0.7, 0.3)	16	31.5	5.1	0	0	47.4
U2 - Beta(1, 1)	16.2	31.4	5.1	0.1	0	47.2
U3 - Beta(0.3, 0.7)	17.3	31	4.1	0.1	0	47.5
U3 - Beta(0.5, 0.5)	17.3	31.9	3.6	0	0	47.2
U3 - Beta(0.7, 0.3)	17.2	31.8	3.8	0	0	47.2
U3 - Beta(1, 1)	18.2	30.7	3.6	0.1	0	47.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	8.2	3.5	0.6	0.1	-
U2 - Beta(0.5, 0.5)	9.4	8.1	3.5	0.6	0.1	-
U2 - Beta(0.7, 0.3)	9.5	8.1	3.4	0.6	0.1	-
U2 - Beta(1, 1)	9.5	8.1	3.5	0.7	0.1	-
U3 - Beta(0.3, 0.7)	9.4	8.1	3.5	0.6	0.1	-
U3 - Beta(0.5, 0.5)	9.4	8.2	3.5	0.6	0.1	-
U3 - Beta(0.7, 0.3)	9.5	8.1	3.5	0.6	0.1	-
U3 - Beta(1, 1)	9.5	8.1	3.5	0.6	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	3.4	1	0.2	0	-
U2 - Beta(0.5, 0.5)	2.6	3.4	1	0.2	0	-
U2 - Beta(0.7, 0.3)	2.6	3.4	1	0.2	0	-
U2 - Beta(1, 1)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.3, 0.7)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.5, 0.5)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.7, 0.3)	2.6	3.4	1	0.2	0	-
U3 - Beta(1, 1)	2.6	3.3	1	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						797.2
U2 - Beta(0.5, 0.5)						797.4
U2 - Beta(0.7, 0.3)						796.6
U2 - Beta(1, 1)						797.9
U3 - Beta(0.3, 0.7)						797.9
U3 - Beta(0.5, 0.5)						798.2
U3 - Beta(0.7, 0.3)						797.5
U3 - Beta(1, 1)						797

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	71.8	22.5	2.6	0.1	0	3
U2 - Beta(0.5, 0.5)	71	23	3	0.1	0	2.9
U2 - Beta(0.7, 0.3)	71.7	22.5	2.8	0	0	3
U2 - Beta(1, 1)	71.7	22.4	3	0.1	0	2.8
U3 - Beta(0.3, 0.7)	71.9	22.7	2.3	0.1	0	3
U3 - Beta(0.5, 0.5)	72.8	21.7	2.3	0.2	0	3
U3 - Beta(0.7, 0.3)	72.7	21.6	2.6	0.1	0	3
U3 - Beta(1, 1)	73	21.7	2.2	0.1	0	3
Mean administrations						
U2 - Beta(0.3, 0.7)	18.6	7.3	3.1	0.4	0	-
U2 - Beta(0.5, 0.5)	18.6	7.2	3.1	0.4	0	-
U2 - Beta(0.7, 0.3)	18.7	7.2	3.1	0.4	0	-
U2 - Beta(1, 1)	18.6	7.3	3.1	0.4	0	-
U3 - Beta(0.3, 0.7)	18.5	7.3	3.1	0.4	0	-
U3 - Beta(0.5, 0.5)	18.7	7.2	3.1	0.4	0	-
U3 - Beta(0.7, 0.3)	18.7	7.3	3.1	0.4	0	-
U3 - Beta(1, 1)	18.6	7.3	3.1	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.8	2.4	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	4.8	2.4	0.9	0.1	0	-
U2 - Beta(0.7, 0.3)	4.8	2.4	0.9	0.1	0	-
U2 - Beta(1, 1)	4.8	2.4	0.8	0.1	0	-
U3 - Beta(0.3, 0.7)	4.8	2.4	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	4.8	2.4	0.8	0.1	0	-
U3 - Beta(0.7, 0.3)	4.8	2.4	0.8	0.1	0	-
U3 - Beta(1, 1)	4.8	2.4	0.8	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1170.7
U2 - Beta(0.5, 0.5)						1170.1
U2 - Beta(0.7, 0.3)						1170
U2 - Beta(1, 1)						1171
U3 - Beta(0.3, 0.7)						1171.1
U3 - Beta(0.5, 0.5)						1170.3
U3 - Beta(0.7, 0.3)						1169.9
U3 - Beta(1, 1)						1170.2

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.1	15.1	16.4	36.5	10.3	1.6
U2 - Beta(0.5, 0.5)	18.8	15.5	15.6	37.6	10.9	1.6
U2 - Beta(0.7, 0.3)	20.1	16.3	15.3	36.8	9.9	1.6
U2 - Beta(1, 1)	21.3	16.3	13.1	37.4	10.3	1.6
U3 - Beta(0.3, 0.7)	24.4	17.2	14.6	33.5	8.7	1.6
U3 - Beta(0.5, 0.5)	26.2	17.5	13.8	32.5	8.4	1.6
U3 - Beta(0.7, 0.3)	28.3	18.2	12.6	30.3	8.9	1.7
U3 - Beta(1, 1)	29.5	18.6	12.2	30.2	7.8	1.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	4.5	6	8.3	5.7	-
U2 - Beta(0.5, 0.5)	5.2	4.6	6.2	8.1	5.6	-
U2 - Beta(0.7, 0.3)	5.3	4.7	6.1	8.1	5.5	-
U2 - Beta(1, 1)	5.3	4.7	6	8.2	5.6	-
U3 - Beta(0.3, 0.7)	5.1	4.5	6.2	8.2	5.7	-
U3 - Beta(0.5, 0.5)	5.2	4.6	6.2	8.1	5.7	-
U3 - Beta(0.7, 0.3)	5.3	4.8	6.1	8	5.5	-
U3 - Beta(1, 1)	5.3	4.7	5.9	8.1	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	2.8	1.9	2.9	1.4	-
U2 - Beta(0.5, 0.5)	1	2.8	2	2.8	1.4	-
U2 - Beta(0.7, 0.3)	1	2.8	1.9	2.8	1.4	-
U2 - Beta(1, 1)	1	2.8	1.9	2.8	1.4	-
U3 - Beta(0.3, 0.7)	0.9	2.7	2	2.9	1.4	-
U3 - Beta(0.5, 0.5)	1	2.8	2	2.8	1.4	-
U3 - Beta(0.7, 0.3)	1	2.8	1.9	2.8	1.4	-
U3 - Beta(1, 1)	1	2.8	1.9	2.8	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1092.5
U2 - Beta(0.5, 0.5)						1089.8
U2 - Beta(0.7, 0.3)						1088.7
U2 - Beta(1, 1)						1090.3
U3 - Beta(0.3, 0.7)						1090.6
U3 - Beta(0.5, 0.5)						1090.9
U3 - Beta(0.7, 0.3)						1088.2
U3 - Beta(1, 1)						1089.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.6	34.9	28.1	24.2	6.9	0.3
U2 - Beta(0.5, 0.5)	6	34.7	27	24.5	7.5	0.3
U2 - Beta(0.7, 0.3)	6.2	35.8	26.2	24.8	6.7	0.3
U2 - Beta(1, 1)	6	35.5	27.4	24	6.8	0.3
U3 - Beta(0.3, 0.7)	8.4	38	27	21.8	4.5	0.3
U3 - Beta(0.5, 0.5)	8.9	39.9	26	21	3.9	0.3
U3 - Beta(0.7, 0.3)	9.8	40.7	25.4	20.1	3.7	0.3
U3 - Beta(1, 1)	9.5	41	25.3	20.1	3.8	0.3
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.7	6.4	7.9	5.8	-
U2 - Beta(0.5, 0.5)	4.1	5.7	6.4	7.9	5.8	-
U2 - Beta(0.7, 0.3)	4.1	5.9	6.5	7.8	5.7	-
U2 - Beta(1, 1)	4.1	5.8	6.4	7.9	5.8	-
U3 - Beta(0.3, 0.7)	4.1	5.7	6.4	7.9	5.9	-
U3 - Beta(0.5, 0.5)	4.1	5.7	6.4	7.9	5.7	-
U3 - Beta(0.7, 0.3)	4.1	5.9	6.5	7.8	5.6	-
U3 - Beta(1, 1)	4.1	5.8	6.4	8	5.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	1.7	2	1.5	-
U2 - Beta(0.5, 0.5)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.5	-
U2 - Beta(1, 1)	0.4	2.9	1.6	2	1.5	-
U3 - Beta(0.3, 0.7)	0.5	2.8	1.7	2	1.5	-
U3 - Beta(0.5, 0.5)	0.5	2.8	1.7	2	1.5	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.5	-
U3 - Beta(1, 1)	0.5	2.9	1.6	2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1185.8
U2 - Beta(0.5, 0.5)						1185.1
U2 - Beta(0.7, 0.3)						1183.4
U2 - Beta(1, 1)						1185.3
U3 - Beta(0.3, 0.7)						1186
U3 - Beta(0.5, 0.5)						1184.3
U3 - Beta(0.7, 0.3)						1183.3
U3 - Beta(1, 1)						1185.1
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.2	14.4	34.7	29.3	12.6	0.8
U2 - Beta(0.5, 0.5)	8.4	12.8	37.1	28.4	12.5	0.8
U2 - Beta(0.7, 0.3)	9.9	14.7	35.1	29	10.5	0.8
U2 - Beta(1, 1)	10.3	14.3	36.8	28.4	9.5	0.7
U3 - Beta(0.3, 0.7)	12.8	19.3	33.4	25.2	8.5	0.8
U3 - Beta(0.5, 0.5)	13.5	18.5	36.1	25	6.1	0.8
U3 - Beta(0.7, 0.3)	14.7	18.5	34.5	25.5	6.1	0.7
U3 - Beta(1, 1)	17.2	19.8	34.1	23.2	4.9	0.8
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	4.1	7.7	7.3	6.8	-
U2 - Beta(0.5, 0.5)	3.9	4.3	7.8	7.3	6.5	-
U2 - Beta(0.7, 0.3)	4	4.4	7.6	7.4	6.5	-
U2 - Beta(1, 1)	4	4.4	7.8	7.3	6.5	-
U3 - Beta(0.3, 0.7)	3.9	4.1	7.7	7.3	6.8	-
U3 - Beta(0.5, 0.5)	4	4.3	7.8	7.3	6.5	-
U3 - Beta(0.7, 0.3)	4	4.4	7.7	7.3	6.5	-
U3 - Beta(1, 1)	4	4.4	7.7	7.4	6.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.8	2.3	-
U2 - Beta(0.5, 0.5)	0.4	2.8	2.5	2.8	2.2	-
U2 - Beta(0.7, 0.3)	0.5	2.9	2.4	2.8	2.2	-
U2 - Beta(1, 1)	0.4	2.9	2.5	2.8	2.1	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.7	2.2	-
U3 - Beta(0.5, 0.5)	0.4	2.8	2.5	2.7	2.2	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.8	2.2	-
U3 - Beta(1, 1)	0.4	2.9	2.5	2.8	2.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057.7
U2 - Beta(0.5, 0.5)						1056.2
U2 - Beta(0.7, 0.3)						1054.9
U2 - Beta(1, 1)						1054.6
U3 - Beta(0.3, 0.7)						1057.8
U3 - Beta(0.5, 0.5)						1054.6
U3 - Beta(0.7, 0.3)						1053.5
U3 - Beta(1, 1)						1055.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.3	7.5	13.9	41.3	29.9	0.1
U2 - Beta(0.5, 0.5)	7.9	7.2	13.7	41.9	29.2	0.1
U2 - Beta(0.7, 0.3)	8.3	6.9	13.8	42.7	28.1	0.2
U2 - Beta(1, 1)	9.1	7.4	13.6	40.6	29.1	0.2
U3 - Beta(0.3, 0.7)	11.5	8.6	13	39.2	27.6	0.1
U3 - Beta(0.5, 0.5)	12.4	8.9	14	39.8	24.8	0.1
U3 - Beta(0.7, 0.3)	15.2	9.8	13.1	38.4	23.3	0.2
U3 - Beta(1, 1)	15.9	9.6	12.7	38.8	22.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.6	5.2	8.1	9.7	-
U2 - Beta(0.5, 0.5)	3.5	3.7	5.3	8.2	9.4	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.2	8.2	9.4	-
U2 - Beta(1, 1)	3.5	3.7	5.1	8.2	9.5	-
U3 - Beta(0.3, 0.7)	3.4	3.5	5.2	8.1	9.7	-
U3 - Beta(0.5, 0.5)	3.5	3.6	5.2	8.2	9.4	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.3	8.1	9.3	-
U3 - Beta(1, 1)	3.5	3.7	5.2	8.2	9.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(0.5, 0.5)	0.2	2.7	1.6	3.1	2.4	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.6	3.1	2.4	-
U2 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.4	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.6	3.1	2.3	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.6	3.1	2.3	-
U3 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1107.9
U2 - Beta(0.5, 0.5)						1105.5
U2 - Beta(0.7, 0.3)						1105.5
U2 - Beta(1, 1)						1106.6
U3 - Beta(0.3, 0.7)						1110.1
U3 - Beta(0.5, 0.5)						1106.4
U3 - Beta(0.7, 0.3)						1105.7
U3 - Beta(1, 1)						1106.2
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.7	13.6	49.6	15.7	4.4	5
U2 - Beta(0.5, 0.5)	13.8	12.8	49.5	15.6	3.4	4.9
U2 - Beta(0.7, 0.3)	14	12.4	47.9	16.7	3.7	5.3
U2 - Beta(1, 1)	14.3	13.3	48.5	15.6	3.3	5
U3 - Beta(0.3, 0.7)	17.2	16	44	14.8	3	5
U3 - Beta(0.5, 0.5)	18.6	13.5	46.9	13.5	2.2	5.3
U3 - Beta(0.7, 0.3)	21.6	14.1	42.2	14.6	2.4	5.1
U3 - Beta(1, 1)	24.5	14.2	40.1	13.9	2.2	5.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	5.8	10.1	5.2	3	-
U2 - Beta(0.5, 0.5)	5.1	5.9	10	5.3	2.8	-
U2 - Beta(0.7, 0.3)	5.3	6	9.8	5.2	2.9	-
U2 - Beta(1, 1)	5.3	5.9	10	5.2	2.8	-
U3 - Beta(0.3, 0.7)	5.1	5.8	10.1	5.3	2.9	-
U3 - Beta(0.5, 0.5)	5.2	5.9	10	5.2	2.8	-
U3 - Beta(0.7, 0.3)	5.2	6	9.9	5.3	2.8	-
U3 - Beta(1, 1)	5.3	6	9.7	5.3	2.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.3	3.2	1.8	0.9	-
U2 - Beta(0.5, 0.5)	1	3.3	3.2	1.8	0.9	-
U2 - Beta(0.7, 0.3)	1	3.3	3.1	1.8	0.9	-
U2 - Beta(1, 1)	1	3.3	3.2	1.8	0.9	-
U3 - Beta(0.3, 0.7)	1	3.3	3.2	1.9	0.9	-
U3 - Beta(0.5, 0.5)	1	3.3	3.2	1.8	0.9	-
U3 - Beta(0.7, 0.3)	1	3.4	3.1	1.9	0.8	-
U3 - Beta(1, 1)	1.1	3.3	3.1	1.8	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1045.3
U2 - Beta(0.5, 0.5)						1039.1
U2 - Beta(0.7, 0.3)						1038.9
U2 - Beta(1, 1)						1040.2
U3 - Beta(0.3, 0.7)						1043.5
U3 - Beta(0.5, 0.5)						1039.3
U3 - Beta(0.7, 0.3)						1038.4
U3 - Beta(1, 1)						1039.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.6	41.1	33.7	10.2	3.2	0.2
U2 - Beta(0.5, 0.5)	11	42.5	33	10.8	2.5	0.2
U2 - Beta(0.7, 0.3)	12.4	41.3	32.2	10.7	3.2	0.2
U2 - Beta(1, 1)	12.3	42.4	31.9	10.6	2.6	0.2
U3 - Beta(0.3, 0.7)	14.3	45.4	29.4	9	1.7	0.2
U3 - Beta(0.5, 0.5)	15.8	47.4	26.1	8.8	1.7	0.2
U3 - Beta(0.7, 0.3)	17.7	46.2	26	8	1.9	0.2
U3 - Beta(1, 1)	17.7	47.1	25.2	8.2	1.6	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	8.5	9.1	5.1	2.8	-
U2 - Beta(0.5, 0.5)	4.5	8.8	9	5.1	2.7	-
U2 - Beta(0.7, 0.3)	4.6	8.8	9	5	2.6	-
U2 - Beta(1, 1)	4.5	8.7	9	5.1	2.6	-
U3 - Beta(0.3, 0.7)	4.5	8.5	9.2	5.1	2.7	-
U3 - Beta(0.5, 0.5)	4.5	8.7	9	5.1	2.6	-
U3 - Beta(0.7, 0.3)	4.6	8.9	8.9	4.9	2.6	-
U3 - Beta(1, 1)	4.5	8.8	9	5	2.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.7	2.3	1.5	0.8	-
U2 - Beta(0.5, 0.5)	0.6	3.8	2.2	1.5	0.7	-
U2 - Beta(0.7, 0.3)	0.7	3.8	2.2	1.5	0.7	-
U2 - Beta(1, 1)	0.6	3.8	2.2	1.5	0.7	-
U3 - Beta(0.3, 0.7)	0.6	3.7	2.3	1.5	0.7	-
U3 - Beta(0.5, 0.5)	0.6	3.8	2.3	1.5	0.7	-
U3 - Beta(0.7, 0.3)	0.7	3.8	2.2	1.4	0.7	-
U3 - Beta(1, 1)	0.6	3.8	2.2	1.5	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1165.9
U2 - Beta(0.5, 0.5)						1165.9
U2 - Beta(0.7, 0.3)						1165.2
U2 - Beta(1, 1)						1165.1
U3 - Beta(0.3, 0.7)						1165.7
U3 - Beta(0.5, 0.5)						1165.3
U3 - Beta(0.7, 0.3)						1164.5
U3 - Beta(1, 1)						1164.2
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.2	37.7	13.7	4.7	1.2	27.5
U2 - Beta(0.5, 0.5)	16.5	37.1	12.9	5.3	0.8	27.4
U2 - Beta(0.7, 0.3)	16.8	37.8	12.7	4.4	1	27.3
U2 - Beta(1, 1)	18.4	37.3	11.8	4.4	0.8	27.3
U3 - Beta(0.3, 0.7)	19.3	37.2	11.2	4.2	0.8	27.3
U3 - Beta(0.5, 0.5)	20.6	36.2	10.6	4.3	0.7	27.6
U3 - Beta(0.7, 0.3)	21.3	37.2	9.6	3.7	0.8	27.4
U3 - Beta(1, 1)	23.5	36.4	8.9	3.2	0.7	27.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	8.2	5.5	2.3	1	-
U2 - Beta(0.5, 0.5)	8.3	8.4	5.2	2.3	1	-
U2 - Beta(0.7, 0.3)	8.3	8.4	5.2	2.3	0.9	-
U2 - Beta(1, 1)	8.3	8.4	5.3	2.3	1	-
U3 - Beta(0.3, 0.7)	8.2	8.2	5.4	2.4	1	-
U3 - Beta(0.5, 0.5)	8.3	8.4	5.2	2.4	0.9	-
U3 - Beta(0.7, 0.3)	8.3	8.5	5.2	2.3	1	-
U3 - Beta(1, 1)	8.4	8.4	5.2	2.3	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.9	1.8	0.9	0.3	-
U2 - Beta(0.5, 0.5)	2.2	4	1.7	0.9	0.3	-
U2 - Beta(0.7, 0.3)	2.2	4	1.7	0.9	0.3	-
U2 - Beta(1, 1)	2.2	4	1.7	0.9	0.3	-
U3 - Beta(0.3, 0.7)	2.1	3.9	1.7	0.9	0.3	-
U3 - Beta(0.5, 0.5)	2.2	4	1.7	0.9	0.3	-
U3 - Beta(0.7, 0.3)	2.2	4	1.7	0.9	0.3	-
U3 - Beta(1, 1)	2.2	4	1.7	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						889.9
U2 - Beta(0.5, 0.5)						890.1
U2 - Beta(0.7, 0.3)						890.9
U2 - Beta(1, 1)						891.4
U3 - Beta(0.3, 0.7)						891.4
U3 - Beta(0.5, 0.5)						892.1
U3 - Beta(0.7, 0.3)						892.5
U3 - Beta(1, 1)						891.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.7	47.2	12.2	1.7	0.1	18.1
U2 - Beta(0.5, 0.5)	21	47.2	12	1.5	0.2	18.1
U2 - Beta(0.7, 0.3)	21.7	45.9	12.7	1.7	0.2	17.8
U2 - Beta(1, 1)	22.6	46.7	11.7	1.2	0.2	17.6
U3 - Beta(0.3, 0.7)	24.9	45.9	9.8	1.5	0	17.9
U3 - Beta(0.5, 0.5)	25.4	46	9.4	1.2	0	18
U3 - Beta(0.7, 0.3)	28.6	44.3	8.1	1.4	0	17.6
U3 - Beta(1, 1)	28.8	43.9	8.5	0.8	0	18
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	10.5	6	2	0.7	-
U2 - Beta(0.5, 0.5)	7.7	10.6	5.8	2	0.7	-
U2 - Beta(0.7, 0.3)	7.8	10.6	5.8	1.9	0.7	-
U2 - Beta(1, 1)	7.8	10.6	5.8	1.9	0.7	-
U3 - Beta(0.3, 0.7)	7.7	10.4	6	1.9	0.7	-
U3 - Beta(0.5, 0.5)	7.7	10.6	5.9	1.9	0.7	-
U3 - Beta(0.7, 0.3)	7.8	10.7	5.7	1.9	0.7	-
U3 - Beta(1, 1)	7.8	10.7	5.7	1.9	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	4.9	1.9	0.8	0.2	-
U2 - Beta(0.5, 0.5)	1.9	4.9	1.8	0.8	0.2	-
U2 - Beta(0.7, 0.3)	1.9	5	1.8	0.8	0.2	-
U2 - Beta(1, 1)	1.9	5	1.8	0.8	0.2	-
U3 - Beta(0.3, 0.7)	1.9	4.9	1.9	0.8	0.2	-
U3 - Beta(0.5, 0.5)	1.9	4.9	1.8	0.8	0.2	-
U3 - Beta(0.7, 0.3)	1.9	5	1.8	0.8	0.2	-
U3 - Beta(1, 1)	1.9	5	1.8	0.8	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						939.3
U2 - Beta(0.5, 0.5)						940.9
U2 - Beta(0.7, 0.3)						939.3
U2 - Beta(1, 1)						939.1
U3 - Beta(0.3, 0.7)						938
U3 - Beta(0.5, 0.5)						939.8
U3 - Beta(0.7, 0.3)						938.8
U3 - Beta(1, 1)						938.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.7	8.8	1.8	0.2	0	75.5
U2 - Beta(0.5, 0.5)	15.2	8	1.6	0.4	0.1	74.7
U2 - Beta(0.7, 0.3)	15.1	8.4	1.4	0.3	0.1	74.7
U2 - Beta(1, 1)	14.8	9.2	1.3	0.4	0.1	74.2
U3 - Beta(0.3, 0.7)	13.7	7.7	0.9	0.2	0.1	77.4
U3 - Beta(0.5, 0.5)	15	9.3	1.2	0.1	0	74.4
U3 - Beta(0.7, 0.3)	14.8	9.1	1.1	0.1	0	74.9
U3 - Beta(1, 1)	14.8	9.4	1.1	0.1	0	74.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	5.2	1.9	0.8	0.1	-
U2 - Beta(0.5, 0.5)	11.2	5.3	1.9	1	0.2	-
U2 - Beta(0.7, 0.3)	11.1	5.3	1.9	1	0.2	-
U2 - Beta(1, 1)	11.1	5.3	1.9	1	0.2	-
U3 - Beta(0.3, 0.7)	11.3	5.2	1.8	0.8	0.1	-
U3 - Beta(0.5, 0.5)	11.3	5.2	1.9	1	0.2	-
U3 - Beta(0.7, 0.3)	11.3	5.3	1.9	0.9	0.2	-
U3 - Beta(1, 1)	11.3	5.3	1.8	0.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.7	0.7	0.4	0.1	-
U2 - Beta(0.5, 0.5)	3	2.8	0.6	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3	2.8	0.6	0.4	0.1	-
U2 - Beta(1, 1)	3	2.8	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.1	2.7	0.6	0.4	0	-
U3 - Beta(0.5, 0.5)	3.1	2.7	0.6	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.1	2.8	0.6	0.4	0.1	-
U3 - Beta(1, 1)	3.1	2.8	0.6	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						676.1
U2 - Beta(0.5, 0.5)						688.8
U2 - Beta(0.7, 0.3)						691
U2 - Beta(1, 1)						688.8
U3 - Beta(0.3, 0.7)						672.8
U3 - Beta(0.5, 0.5)						686.5
U3 - Beta(0.7, 0.3)						689.2
U3 - Beta(1, 1)						687.8
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.1	29.3	35.5	3.6	0.4	3.1
U2 - Beta(0.5, 0.5)	26.9	19.4	38.3	11.5	1	2.9
U2 - Beta(0.7, 0.3)	26.2	23	33.2	13.1	1.9	2.6
U2 - Beta(1, 1)	26	20.2	37.6	12.5	0.9	2.8
U3 - Beta(0.3, 0.7)	36.1	33.6	24.4	2.5	0.1	3.3
U3 - Beta(0.5, 0.5)	31.1	19.8	35.2	10.2	0.9	2.8
U3 - Beta(0.7, 0.3)	28.2	22.2	32.8	12.4	1.2	3.2
U3 - Beta(1, 1)	30.9	20	35.8	9.7	0.8	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	12.4	9.8	6.6	0.8	0	-
U2 - Beta(0.5, 0.5)	9.6	7.6	8.8	3.1	0.5	-
U2 - Beta(0.7, 0.3)	9.7	7.2	7.5	4.2	1	-
U2 - Beta(1, 1)	9.5	7.6	8.7	3.2	0.5	-
U3 - Beta(0.3, 0.7)	13.7	10.2	5.1	0.5	0	-
U3 - Beta(0.5, 0.5)	10.1	7.7	8.5	2.8	0.4	-
U3 - Beta(0.7, 0.3)	9.9	7.4	7.8	3.9	0.7	-
U3 - Beta(1, 1)	10.2	7.8	8.5	2.8	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	4.1	2	0.2	0	-
U2 - Beta(0.5, 0.5)	2.4	3.4	2.4	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.3	2	1.2	0.3	-
U2 - Beta(1, 1)	2.4	3.4	2.4	1	0.1	-
U3 - Beta(0.3, 0.7)	4.2	4.3	1.5	0.2	0	-
U3 - Beta(0.5, 0.5)	2.6	3.5	2.4	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.4	2.1	1.1	0.2	-
U3 - Beta(1, 1)	2.6	3.5	2.3	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1077.5
U2 - Beta(0.5, 0.5)						1113
U2 - Beta(0.7, 0.3)						1120
U2 - Beta(1, 1)						1116.1
U3 - Beta(0.3, 0.7)						1064.9
U3 - Beta(0.5, 0.5)						1111.1
U3 - Beta(0.7, 0.3)						1116.5
U3 - Beta(1, 1)						1109.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	65	16.6	3.4	0	9
U2 - Beta(0.5, 0.5)	8	47	24.8	9.7	1.3	9.2
U2 - Beta(0.7, 0.3)	7.7	50.4	22.9	8.6	1.5	8.9
U2 - Beta(1, 1)	8.1	47.1	24.7	9.5	1.4	9.2
U3 - Beta(0.3, 0.7)	7	68.8	13	2.1	0	9.1
U3 - Beta(0.5, 0.5)	9.2	51.6	21.9	7.2	1	9.1
U3 - Beta(0.7, 0.3)	9.5	51.1	20.3	8.5	1.6	9
U3 - Beta(1, 1)	8.4	54	21.1	6.4	0.9	9.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	16.7	3.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	6.8	11.6	6.6	3	0.7	-
U2 - Beta(0.7, 0.3)	6.7	10.2	6.6	4	1.2	-
U2 - Beta(1, 1)	6.7	11.8	6.5	3	0.7	-
U3 - Beta(0.3, 0.7)	8.1	17	3	0.5	0.1	-
U3 - Beta(0.5, 0.5)	6.9	12.4	6.1	2.7	0.5	-
U3 - Beta(0.7, 0.3)	6.9	11	6.3	3.6	0.9	-
U3 - Beta(1, 1)	6.9	12.7	6	2.7	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	6.4	1.2	0.2	0	-
U2 - Beta(0.5, 0.5)	1.3	4.7	2	1	0.2	-
U2 - Beta(0.7, 0.3)	1.3	4.4	1.8	1.3	0.4	-
U2 - Beta(1, 1)	1.2	4.8	2	1	0.2	-
U3 - Beta(0.3, 0.7)	1.9	6.5	1	0.2	0	-
U3 - Beta(0.5, 0.5)	1.4	5	1.8	0.9	0.2	-
U3 - Beta(0.7, 0.3)	1.3	4.6	1.9	1.1	0.3	-
U3 - Beta(1, 1)	1.4	5.1	1.8	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1054
U2 - Beta(0.5, 0.5)						1070
U2 - Beta(0.7, 0.3)						1076.4
U2 - Beta(1, 1)						1069.1
U3 - Beta(0.3, 0.7)						1048.9
U3 - Beta(0.5, 0.5)						1065.9
U3 - Beta(0.7, 0.3)						1071.2
U3 - Beta(1, 1)						1065.6
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.8	6.2	31.6	53.3	3.1	2
U2 - Beta(0.5, 0.5)	6.7	5.4	18.8	55.1	12.4	1.6
U2 - Beta(0.7, 0.3)	7.3	6.4	20.2	50.2	14.2	1.7
U2 - Beta(1, 1)	6.6	5.7	19.4	54.7	12	1.6
U3 - Beta(0.3, 0.7)	5.7	8.2	32.1	50.4	1.8	1.8
U3 - Beta(0.5, 0.5)	8.6	6.8	19.4	54	9.6	1.6
U3 - Beta(0.7, 0.3)	8.2	8.4	19.2	49.6	12.9	1.7
U3 - Beta(1, 1)	9.2	7.9	16.8	55.9	8.4	1.8
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	5.5	10.4	8.8	0.6	-
U2 - Beta(0.5, 0.5)	4.2	4.7	7.1	10.7	3.1	-
U2 - Beta(0.7, 0.3)	4.2	4.7	6.9	9.7	4.3	-
U2 - Beta(1, 1)	4.2	4.8	7.1	10.8	3	-
U3 - Beta(0.3, 0.7)	4.7	5.8	10.4	8.4	0.4	-
U3 - Beta(0.5, 0.5)	4.4	4.8	7.2	10.8	2.7	-
U3 - Beta(0.7, 0.3)	4.5	5	6.9	9.7	3.7	-
U3 - Beta(1, 1)	4.6	5	7	10.7	2.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.2	3.7	2.9	0.2	-
U2 - Beta(0.5, 0.5)	0.5	3.1	2.1	3.4	0.9	-
U2 - Beta(0.7, 0.3)	0.5	3.1	2	3.1	1.1	-
U2 - Beta(1, 1)	0.5	3.1	2.1	3.4	0.8	-
U3 - Beta(0.3, 0.7)	0.8	3.3	3.7	2.8	0.1	-
U3 - Beta(0.5, 0.5)	0.6	3.1	2.1	3.4	0.7	-
U3 - Beta(0.7, 0.3)	0.7	3.2	1.9	3.1	1	-
U3 - Beta(1, 1)	0.7	3.2	2	3.4	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1055.6
U2 - Beta(0.5, 0.5)						1098.6
U2 - Beta(0.7, 0.3)						1102.9
U2 - Beta(1, 1)						1097.3
U3 - Beta(0.3, 0.7)						1050.8
U3 - Beta(0.5, 0.5)						1093.2
U3 - Beta(0.7, 0.3)						1095.9
U3 - Beta(1, 1)						1090.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.9	43.4	28.9	19.4	5.7	1.7
U2 - Beta(0.5, 0.5)	5.2	33.4	26.4	23.1	10.4	1.5
U2 - Beta(0.7, 0.3)	5.9	37.7	26.4	20.3	8.2	1.5
U2 - Beta(1, 1)	6	36.7	25.5	21	9.3	1.5
U3 - Beta(0.3, 0.7)	2.1	52	26.6	14.2	3.5	1.6
U3 - Beta(0.5, 0.5)	6.9	44.3	23.2	18.4	5.7	1.5
U3 - Beta(0.7, 0.3)	7.1	43.3	25.2	16.4	6.5	1.5
U3 - Beta(1, 1)	9.2	45.8	22.9	15.4	5.2	1.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	13.9	7	3.6	1	-
U2 - Beta(0.5, 0.5)	4	7.5	7.2	7.2	3.9	-
U2 - Beta(0.7, 0.3)	4	6.9	7.2	7.4	4.4	-
U2 - Beta(1, 1)	3.9	7.6	7.2	7.1	3.8	-
U3 - Beta(0.3, 0.7)	4.5	14.9	6.6	3	0.7	-
U3 - Beta(0.5, 0.5)	4.1	8.2	7.4	6.8	3.3	-
U3 - Beta(0.7, 0.3)	4.1	7.6	7.1	7.1	3.8	-
U3 - Beta(1, 1)	4.1	8.6	7.3	6.7	3.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	6.1	2.6	1.3	0.4	-
U2 - Beta(0.5, 0.5)	0.3	3.9	2.3	2.7	1.4	-
U2 - Beta(0.7, 0.3)	0.4	3.8	2.2	2.8	1.5	-
U2 - Beta(1, 1)	0.3	4	2.3	2.7	1.4	-
U3 - Beta(0.3, 0.7)	0.7	6.5	2.5	1.1	0.2	-
U3 - Beta(0.5, 0.5)	0.4	4.2	2.4	2.6	1.2	-
U3 - Beta(0.7, 0.3)	0.4	4	2.3	2.6	1.4	-
U3 - Beta(1, 1)	0.4	4.3	2.4	2.5	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1031.9
U2 - Beta(0.5, 0.5)						1047.9
U2 - Beta(0.7, 0.3)						1050.9
U2 - Beta(1, 1)						1048.8
U3 - Beta(0.3, 0.7)						1029.6
U3 - Beta(0.5, 0.5)						1046.7
U3 - Beta(0.7, 0.3)						1048.3
U3 - Beta(1, 1)						1045.2

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	67.7	11.3	2.6	0.4	0	18
U2 - Beta(0.5, 0.5)	66	12.8	3.6	1.1	0.2	16.3
U2 - Beta(0.7, 0.3)	63.5	15	3.2	1.2	0.2	16.9
U2 - Beta(1, 1)	64.8	14	3.3	1.1	0.2	16.6
U3 - Beta(0.3, 0.7)	69.7	10.2	1.9	0.2	0	18
U3 - Beta(0.5, 0.5)	66.4	12.4	3	1.1	0.2	16.9
U3 - Beta(0.7, 0.3)	66.5	13.9	2.7	0.5	0.1	16.3
U3 - Beta(1, 1)	67	13.3	2.5	0.8	0.1	16.3
Mean administrations						
U2 - Beta(0.3, 0.7)	21.8	4.1	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	18.3	5.7	2.2	0.8	0.2	-
U2 - Beta(0.7, 0.3)	17.7	6.1	2.1	1.1	0.3	-
U2 - Beta(1, 1)	18.1	5.9	2.2	0.8	0.2	-
U3 - Beta(0.3, 0.7)	22.3	3.9	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	18.7	5.6	1.9	0.7	0.2	-
U3 - Beta(0.7, 0.3)	18.1	5.9	2.1	1	0.2	-
U3 - Beta(1, 1)	18.9	5.6	1.9	0.7	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6	1.7	0.3	0	0	-
U2 - Beta(0.5, 0.5)	4.7	2.3	0.7	0.3	0.1	-
U2 - Beta(0.7, 0.3)	4.6	2.4	0.6	0.4	0.1	-
U2 - Beta(1, 1)	4.6	2.4	0.7	0.3	0.1	-
U3 - Beta(0.3, 0.7)	6.2	1.6	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.9	2.3	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	4.6	2.4	0.6	0.3	0.1	-
U3 - Beta(1, 1)	4.9	2.3	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.7
U2 - Beta(0.5, 0.5)						1059
U2 - Beta(0.7, 0.3)						1061.6
U2 - Beta(1, 1)						1057.9
U3 - Beta(0.3, 0.7)						1036.6
U3 - Beta(0.5, 0.5)						1054.9
U3 - Beta(0.7, 0.3)						1061.9
U3 - Beta(1, 1)						1059.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.6	71.4	7.9	0.7	0.1	3.3
U2 - Beta(0.5, 0.5)	12.4	51	24.4	7.8	1.5	2.9
U2 - Beta(0.7, 0.3)	13.6	46.4	22.6	12.5	2.1	2.8
U2 - Beta(1, 1)	11.5	51.1	24.4	8.4	1.6	3
U3 - Beta(0.3, 0.7)	19.6	71.5	4.9	0.6	0	3.4
U3 - Beta(0.5, 0.5)	14.2	55.8	19	6.8	1.2	3
U3 - Beta(0.7, 0.3)	14.4	48.6	23	9.8	1.4	2.8
U3 - Beta(1, 1)	13.8	57.2	18.6	6.5	0.8	3.1
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	16.2	1.6	0.2	0	-
U2 - Beta(0.5, 0.5)	7.9	13.7	5.6	2	0.4	-
U2 - Beta(0.7, 0.3)	7.7	10.1	6.3	4.1	1.4	-
U2 - Beta(1, 1)	7.8	13.5	5.7	2.1	0.4	-
U3 - Beta(0.3, 0.7)	12	16.2	1.1	0.2	0	-
U3 - Beta(0.5, 0.5)	8.2	14.2	5	1.8	0.3	-
U3 - Beta(0.7, 0.3)	7.9	11.4	6.2	3.4	0.8	-
U3 - Beta(1, 1)	8.2	14.4	4.9	1.7	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	5.5	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	1.6	4.9	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	1.6	3.9	1.6	1.2	0.4	-
U2 - Beta(1, 1)	1.6	4.9	1.6	0.7	0.1	-
U3 - Beta(0.3, 0.7)	3.4	5.5	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	1.7	5.1	1.3	0.6	0.1	-
U3 - Beta(0.7, 0.3)	1.6	4.2	1.7	1	0.2	-
U3 - Beta(1, 1)	1.7	5.2	1.3	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1118.8
U2 - Beta(0.5, 0.5)						1145.7
U2 - Beta(0.7, 0.3)						1153
U2 - Beta(1, 1)						1146
U3 - Beta(0.3, 0.7)						1115.7
U3 - Beta(0.5, 0.5)						1143.1
U3 - Beta(0.7, 0.3)						1149.2
U3 - Beta(1, 1)						1140.8
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.5	8.3	3	0.7	0.1	61.4
U2 - Beta(0.5, 0.5)	29.4	8.3	1.7	0.4	0.2	60
U2 - Beta(0.7, 0.3)	27.7	10.1	1.7	0.4	0.2	59.9
U2 - Beta(1, 1)	28	8.9	1.9	0.5	0.2	60.5
U3 - Beta(0.3, 0.7)	27.9	7.3	1.9	0.6	0	62.3
U3 - Beta(0.5, 0.5)	29.4	8.1	1.6	0.3	0.2	60.4
U3 - Beta(0.7, 0.3)	29.3	8.8	1.6	0.4	0.2	59.7
U3 - Beta(1, 1)	29.4	8.3	1.1	0.4	0.2	60.6
Mean administrations						
U2 - Beta(0.3, 0.7)	13.9	5.2	1.5	0.4	0	-
U2 - Beta(0.5, 0.5)	13.4	5.3	1.8	0.8	0.1	-
U2 - Beta(0.7, 0.3)	13.4	5.3	1.7	0.8	0.2	-
U2 - Beta(1, 1)	13.3	5.4	1.8	0.8	0.1	-
U3 - Beta(0.3, 0.7)	14.3	5	1.2	0.3	0	-
U3 - Beta(0.5, 0.5)	13.6	5.2	1.6	0.7	0.1	-
U3 - Beta(0.7, 0.3)	13.5	5.4	1.6	0.8	0.1	-
U3 - Beta(1, 1)	13.5	5.3	1.6	0.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	2.5	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.6	2.6	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.7	2.6	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.6	2.6	0.6	0.3	0	-
U3 - Beta(0.3, 0.7)	4	2.5	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	3.7	2.5	0.5	0.3	0	-
U3 - Beta(0.7, 0.3)	3.7	2.6	0.5	0.3	0	-
U3 - Beta(1, 1)	3.7	2.6	0.5	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						757.1
U2 - Beta(0.5, 0.5)						775.8
U2 - Beta(0.7, 0.3)						779.5
U2 - Beta(1, 1)						777.8
U3 - Beta(0.3, 0.7)						755.4
U3 - Beta(0.5, 0.5)						773.1
U3 - Beta(0.7, 0.3)						776.6
U3 - Beta(1, 1)						774.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	10.9	25.6	22.7	15	17.7
U2 - Beta(0.5, 0.5)	11.6	11.7	21.5	17.5	21.8	15.9
U2 - Beta(0.7, 0.3)	11.2	12.9	21.5	15.5	22.2	16.7
U2 - Beta(1, 1)	13.5	10.7	21.2	16.1	22.1	16.4
U3 - Beta(0.3, 0.7)	12.5	14.6	23.8	19.9	10.8	18.4
U3 - Beta(0.5, 0.5)	15.5	13.9	19.7	16	17.8	17.1
U3 - Beta(0.7, 0.3)	17.4	14.3	18.2	16	17.5	16.6
U3 - Beta(1, 1)	19.3	14.5	18.3	13.9	17.3	16.7
Mean administrations						
U2 - Beta(0.3, 0.7)	7	6.3	7.3	5.1	2.1	-
U2 - Beta(0.5, 0.5)	6.7	5.8	6	5.2	4.2	-
U2 - Beta(0.7, 0.3)	6.7	5.9	5.9	5	4.5	-
U2 - Beta(1, 1)	6.6	6	6.1	5.2	4.2	-
U3 - Beta(0.3, 0.7)	7.5	6.4	7.2	4.9	1.7	-
U3 - Beta(0.5, 0.5)	6.9	5.9	6	5.2	3.9	-
U3 - Beta(0.7, 0.3)	6.9	6.1	5.8	5	4.1	-
U3 - Beta(1, 1)	7.2	6.2	5.7	5	3.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	3.4	2.7	2	0.7	-
U2 - Beta(0.5, 0.5)	1.5	3.4	2	2.1	1.3	-
U2 - Beta(0.7, 0.3)	1.4	3.4	2	2	1.3	-
U2 - Beta(1, 1)	1.4	3.4	2	2	1.3	-
U3 - Beta(0.3, 0.7)	1.9	3.4	2.6	1.9	0.6	-
U3 - Beta(0.5, 0.5)	1.6	3.4	2	2	1.2	-
U3 - Beta(0.7, 0.3)	1.6	3.5	1.9	2	1.2	-
U3 - Beta(1, 1)	1.7	3.6	1.8	2	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						955
U2 - Beta(0.5, 0.5)						977.8
U2 - Beta(0.7, 0.3)						980.3
U2 - Beta(1, 1)						979.4
U3 - Beta(0.3, 0.7)						948.2
U3 - Beta(0.5, 0.5)						974.9
U3 - Beta(0.7, 0.3)						975.8
U3 - Beta(1, 1)						973
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.7	39.3	8.3	1.5	0	44.2
U2 - Beta(0.5, 0.5)	7.6	31.5	14.2	4	0.9	41.8
U2 - Beta(0.7, 0.3)	7.9	32.9	12.7	4.9	1.2	40.4
U2 - Beta(1, 1)	6.9	32.6	14.6	3.5	0.8	41.6
U3 - Beta(0.3, 0.7)	6.2	40.7	7.1	1.2	0	44.8
U3 - Beta(0.5, 0.5)	8.1	33.7	12.5	3.4	0.8	41.5
U3 - Beta(0.7, 0.3)	7.8	33.8	12.8	3.4	0.9	41.3
U3 - Beta(1, 1)	7.3	34.8	12.3	2.9	0.8	41.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.1	12.1	2.2	0.3	0	-
U2 - Beta(0.5, 0.5)	8.6	9.6	4	1.6	0.4	-
U2 - Beta(0.7, 0.3)	8.5	8.7	4.3	2.3	0.6	-
U2 - Beta(1, 1)	8.6	9.6	4	1.6	0.4	-
U3 - Beta(0.3, 0.7)	9.2	12.3	1.9	0.3	0	-
U3 - Beta(0.5, 0.5)	8.7	9.9	3.8	1.5	0.3	-
U3 - Beta(0.7, 0.3)	8.6	9.1	4.1	2	0.5	-
U3 - Beta(1, 1)	8.7	9.9	3.8	1.5	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	5	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.8	4.1	1.1	0.5	0.1	-
U2 - Beta(0.7, 0.3)	1.8	3.9	1.1	0.7	0.2	-
U2 - Beta(1, 1)	1.8	4.1	1.1	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.1	5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	1.9	4.2	1.1	0.5	0.1	-
U3 - Beta(0.7, 0.3)	1.9	4	1.1	0.6	0.1	-
U3 - Beta(1, 1)	1.9	4.3	1.1	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						876.8
U2 - Beta(0.5, 0.5)						903.3
U2 - Beta(0.7, 0.3)						914.9
U2 - Beta(1, 1)						902
U3 - Beta(0.3, 0.7)						873.3
U3 - Beta(0.5, 0.5)						900.6
U3 - Beta(0.7, 0.3)						906.4
U3 - Beta(1, 1)						901.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	59	21.6	4.9	0.1	0	14.4
U2 - Beta(0.5, 0.5)	60.9	21.5	3.3	0.1	0	14.2
U2 - Beta(0.7, 0.3)	59.9	23	2.4	0.3	0	14.4
U2 - Beta(1, 1)	59.1	22.9	4	0.1	0	13.9
U3 - Beta(0.3, 0.7)	61.7	22	2	0	0	14.3
U3 - Beta(0.5, 0.5)	62.6	20.6	2.4	0.1	0	14.3
U3 - Beta(0.7, 0.3)	61.7	20.7	3.1	0.1	0	14.4
U3 - Beta(1, 1)	62	20.5	2.6	0	0	14.9
Mean administrations						
U2 - Beta(0.3, 0.7)	17.8	7.6	2.2	0.4	0	-
U2 - Beta(0.5, 0.5)	16.8	7.5	2.7	1	0.1	-
U2 - Beta(0.7, 0.3)	16.8	7.7	2.6	1	0.1	-
U2 - Beta(1, 1)	16.5	7.8	2.8	1	0.1	-
U3 - Beta(0.3, 0.7)	18.8	7.3	1.6	0.3	0	-
U3 - Beta(0.5, 0.5)	17.2	7.4	2.5	0.9	0.1	-
U3 - Beta(0.7, 0.3)	17	7.5	2.5	1	0.1	-
U3 - Beta(1, 1)	17.2	7.5	2.4	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.4	3.4	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	4.9	3.4	0.8	0.4	0	-
U2 - Beta(0.7, 0.3)	4.9	3.5	0.8	0.4	0	-
U2 - Beta(1, 1)	4.8	3.5	0.9	0.4	0	-
U3 - Beta(0.3, 0.7)	5.8	3.2	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	5	3.4	0.8	0.3	0	-
U3 - Beta(0.7, 0.3)	5	3.5	0.7	0.4	0	-
U3 - Beta(1, 1)	5	3.5	0.7	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1009.4
U2 - Beta(0.5, 0.5)						1022.3
U2 - Beta(0.7, 0.3)						1025.2
U2 - Beta(1, 1)						1025.9
U3 - Beta(0.3, 0.7)						1008.4
U3 - Beta(0.5, 0.5)						1020.2
U3 - Beta(0.7, 0.3)						1023.4
U3 - Beta(1, 1)						1019.5

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	10.3	11.2	59.4	8.2	2.8
U2 - Beta(0.5, 0.5)	11.4	10.8	11.9	47	16.4	2.5
U2 - Beta(0.7, 0.3)	12	13.6	10.7	43.5	17.7	2.5
U2 - Beta(1, 1)	12.1	11.7	10.6	46	17.2	2.4
U3 - Beta(0.3, 0.7)	12.6	13.4	8.7	56	6.3	3
U3 - Beta(0.5, 0.5)	15.3	13.8	9.2	43.7	15.5	2.5
U3 - Beta(0.7, 0.3)	16.2	15.6	9.5	40.8	15.5	2.4
U3 - Beta(1, 1)	16.4	14.1	9.4	42.8	14.8	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	6	6.4	11	1.7	-
U2 - Beta(0.5, 0.5)	4.6	5.3	5.9	9.5	4.5	-
U2 - Beta(0.7, 0.3)	4.7	5.4	5.8	8.5	5.4	-
U2 - Beta(1, 1)	4.6	5.4	5.8	9.4	4.5	-
U3 - Beta(0.3, 0.7)	4.9	6.4	6.3	10.6	1.5	-
U3 - Beta(0.5, 0.5)	4.8	5.4	5.8	9.5	4.2	-
U3 - Beta(0.7, 0.3)	5	5.7	5.6	8.7	4.8	-
U3 - Beta(1, 1)	5.1	5.7	5.6	9.3	4.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	3.4	2.3	3.8	0.5	-
U2 - Beta(0.5, 0.5)	0.7	3.3	1.7	3.4	1.2	-
U2 - Beta(0.7, 0.3)	0.7	3.4	1.7	3.1	1.4	-
U2 - Beta(1, 1)	0.7	3.4	1.7	3.3	1.2	-
U3 - Beta(0.3, 0.7)	0.9	3.6	2.2	3.7	0.4	-
U3 - Beta(0.5, 0.5)	0.8	3.4	1.7	3.4	1.1	-
U3 - Beta(0.7, 0.3)	0.9	3.5	1.6	3.1	1.3	-
U3 - Beta(1, 1)	0.9	3.5	1.6	3.3	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1045
U2 - Beta(0.5, 0.5)						1071.1
U2 - Beta(0.7, 0.3)						1075.8
U2 - Beta(1, 1)						1070.3
U3 - Beta(0.3, 0.7)						1037.4
U3 - Beta(0.5, 0.5)						1068.5
U3 - Beta(0.7, 0.3)						1067.2
U3 - Beta(1, 1)						1064.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	33.8	13.5	3.6	0.1	41.3
U2 - Beta(0.5, 0.5)	9	32	14.2	4.8	1.2	38.8
U2 - Beta(0.7, 0.3)	9	33.9	13.8	4.2	1.2	37.9
U2 - Beta(1, 1)	9.8	31.9	13.9	4.6	1.1	38.7
U3 - Beta(0.3, 0.7)	8.1	36.2	10.7	2.8	0	42.2
U3 - Beta(0.5, 0.5)	10.5	32.6	13.8	3.5	0.8	38.8
U3 - Beta(0.7, 0.3)	10	34.8	12.7	3.8	0.8	37.9
U3 - Beta(1, 1)	11.8	33.8	11.6	3.3	0.9	38.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9	11	3.3	0.9	0.1	-
U2 - Beta(0.5, 0.5)	8.6	8.8	4.5	2.3	0.6	-
U2 - Beta(0.7, 0.3)	8.5	8.7	4.4	2.5	0.8	-
U2 - Beta(1, 1)	8.5	8.9	4.5	2.3	0.6	-
U3 - Beta(0.3, 0.7)	9.2	11.2	2.9	0.8	0	-
U3 - Beta(0.5, 0.5)	8.7	9.1	4.3	2.1	0.5	-
U3 - Beta(0.7, 0.3)	8.7	8.8	4.2	2.4	0.6	-
U3 - Beta(1, 1)	8.7	9.2	4.3	2.1	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	4.8	1.1	0.3	0	-
U2 - Beta(0.5, 0.5)	1.9	4.1	1.4	0.8	0.2	-
U2 - Beta(0.7, 0.3)	1.9	4.1	1.3	0.9	0.2	-
U2 - Beta(1, 1)	1.9	4.1	1.4	0.8	0.2	-
U3 - Beta(0.3, 0.7)	2.2	4.9	1	0.3	0	-
U3 - Beta(0.5, 0.5)	2	4.2	1.3	0.8	0.2	-
U3 - Beta(0.7, 0.3)	2	4.1	1.3	0.8	0.2	-
U3 - Beta(1, 1)	2	4.2	1.3	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						870.9
U2 - Beta(0.5, 0.5)						894.6
U2 - Beta(0.7, 0.3)						901.7
U2 - Beta(1, 1)						895.4
U3 - Beta(0.3, 0.7)						863.4
U3 - Beta(0.5, 0.5)						890.5
U3 - Beta(0.7, 0.3)						896.1
U3 - Beta(1, 1)						891.9
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.9	16.4	16.7	38.7	10.4	6.9
U2 - Beta(0.5, 0.5)	16.3	17.2	15.7	31.7	12.3	6.8
U2 - Beta(0.7, 0.3)	17.3	17.1	17.9	29	11.8	6.9
U2 - Beta(1, 1)	17.7	15.8	17.1	30.4	11.9	7.1
U3 - Beta(0.3, 0.7)	16.4	18.7	13.5	36.1	7.8	7.5
U3 - Beta(0.5, 0.5)	23	20.1	14.6	27	8.3	7
U3 - Beta(0.7, 0.3)	23.6	21.1	13.7	26.5	8.4	6.7
U3 - Beta(1, 1)	24.6	19.8	14	26.5	7.9	7.2
Mean administrations						
U2 - Beta(0.3, 0.7)	6.1	6.8	6.5	8	2	-
U2 - Beta(0.5, 0.5)	6	6.4	5.9	7.4	3.6	-
U2 - Beta(0.7, 0.3)	6	6.4	6	7.1	3.9	-
U2 - Beta(1, 1)	6	6.4	6	7.3	3.7	-
U3 - Beta(0.3, 0.7)	6.6	7	6.3	7.6	1.8	-
U3 - Beta(0.5, 0.5)	6.4	6.5	5.8	7.3	3.4	-
U3 - Beta(0.7, 0.3)	6.4	6.6	5.7	7.1	3.6	-
U3 - Beta(1, 1)	6.6	6.7	5.6	7.2	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.7	2.3	3	0.7	-
U2 - Beta(0.5, 0.5)	1.3	3.7	1.9	2.8	1.2	-
U2 - Beta(0.7, 0.3)	1.3	3.7	1.9	2.7	1.2	-
U2 - Beta(1, 1)	1.3	3.8	1.9	2.8	1.2	-
U3 - Beta(0.3, 0.7)	1.6	3.8	2.3	2.9	0.6	-
U3 - Beta(0.5, 0.5)	1.4	3.8	1.8	2.8	1.1	-
U3 - Beta(0.7, 0.3)	1.5	3.8	1.8	2.7	1.1	-
U3 - Beta(1, 1)	1.5	3.9	1.8	2.7	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						999.9
U2 - Beta(0.5, 0.5)						1014.7
U2 - Beta(0.7, 0.3)						1017.5
U2 - Beta(1, 1)						1015.6
U3 - Beta(0.3, 0.7)						994
U3 - Beta(0.5, 0.5)						1014.2
U3 - Beta(0.7, 0.3)						1013.7
U3 - Beta(1, 1)						1012.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	38.4	30.7	12.3	5.3	0.5	12.8
U2 - Beta(0.5, 0.5)	41.2	29	11.8	4.9	0.5	12.6
U2 - Beta(0.7, 0.3)	41.1	27.9	12.6	4.8	0.7	12.9
U2 - Beta(1, 1)	41.7	26.5	12.9	5.8	0.4	12.7
U3 - Beta(0.3, 0.7)	44	31.4	8.3	3.1	0.2	13
U3 - Beta(0.5, 0.5)	45.7	30.6	7.3	3.5	0.1	12.8
U3 - Beta(0.7, 0.3)	47.4	27.4	8.5	3.2	0.3	13.2
U3 - Beta(1, 1)	46.5	27.2	9.4	3.4	0.3	13.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.7	8.7	5.9	3.4	0.8	-
U2 - Beta(0.5, 0.5)	9.7	8.5	5.8	3.6	1.2	-
U2 - Beta(0.7, 0.3)	9.8	8.3	5.7	3.6	1.2	-
U2 - Beta(1, 1)	9.7	8.5	5.8	3.6	1.1	-
U3 - Beta(0.3, 0.7)	10.7	8.8	5.2	3	0.8	-
U3 - Beta(0.5, 0.5)	10.4	8.3	5.5	3.5	1.1	-
U3 - Beta(0.7, 0.3)	10.4	8.4	5.4	3.4	1	-
U3 - Beta(1, 1)	10.5	8.4	5.3	3.5	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.2	2.2	1.4	0.4	-
U2 - Beta(0.5, 0.5)	2.8	4.3	2	1.5	0.4	-
U2 - Beta(0.7, 0.3)	2.8	4.3	1.9	1.6	0.4	-
U2 - Beta(1, 1)	2.7	4.3	2	1.5	0.4	-
U3 - Beta(0.3, 0.7)	3.3	4.3	1.9	1.2	0.3	-
U3 - Beta(0.5, 0.5)	3.1	4.2	1.8	1.5	0.4	-
U3 - Beta(0.7, 0.3)	3.1	4.3	1.8	1.5	0.4	-
U3 - Beta(1, 1)	3.1	4.3	1.8	1.5	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						966.6
U2 - Beta(0.5, 0.5)						972.4
U2 - Beta(0.7, 0.3)						971.3
U2 - Beta(1, 1)						970.6
U3 - Beta(0.3, 0.7)						961.6
U3 - Beta(0.5, 0.5)						971
U3 - Beta(0.7, 0.3)						969.9
U3 - Beta(1, 1)						969.3
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.2	23.3	20.9	22	13.6	0
U2 - Beta(0.5, 0.5)	37.8	27.2	16	12.1	6.9	0
U2 - Beta(0.7, 0.3)	37.9	29.9	14.1	12.1	6	0
U2 - Beta(1, 1)	39	28.5	15.3	10.7	6.5	0
U3 - Beta(0.3, 0.7)	28.3	26.4	19.1	18	8.2	0
U3 - Beta(0.5, 0.5)	42	30	13.7	10.2	4.1	0
U3 - Beta(0.7, 0.3)	42.3	32	13.2	8.8	3.7	0
U3 - Beta(1, 1)	46.6	29.3	13.4	7.1	3.6	0
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	7.6	6	5.3	2.8	-
U2 - Beta(0.5, 0.5)	4.3	5	5.9	8.4	6.3	-
U2 - Beta(0.7, 0.3)	4.3	5	6.1	8.3	6.4	-
U2 - Beta(1, 1)	4.5	5.2	6.1	8.2	6.1	-
U3 - Beta(0.3, 0.7)	9	8.1	5.8	4.8	2.3	-
U3 - Beta(0.5, 0.5)	4.7	5.4	6.2	8.1	5.7	-
U3 - Beta(0.7, 0.3)	4.8	5.4	6.3	8	5.5	-
U3 - Beta(1, 1)	4.8	5.7	6.4	7.9	5.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	3.7	2.5	2.2	1.2	-
U2 - Beta(0.5, 0.5)	0.5	2.9	2.3	3.6	2.4	-
U2 - Beta(0.7, 0.3)	0.5	2.9	2.2	3.7	2.3	-
U2 - Beta(1, 1)	0.6	2.9	2.3	3.5	2.3	-
U3 - Beta(0.3, 0.7)	2.7	3.9	2.4	2	1	-
U3 - Beta(0.5, 0.5)	0.7	3.1	2.3	3.5	2.1	-
U3 - Beta(0.7, 0.3)	0.8	3.1	2.3	3.5	2	-
U3 - Beta(1, 1)	0.7	3.2	2.4	3.4	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						992.2
U2 - Beta(0.5, 0.5)						1009.1
U2 - Beta(0.7, 0.3)						1012
U2 - Beta(1, 1)						1007.7
U3 - Beta(0.3, 0.7)						990.4
U3 - Beta(0.5, 0.5)						1007.1
U3 - Beta(0.7, 0.3)						1007.8
U3 - Beta(1, 1)						1006.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.1	55.2	4	0.2	0	0.5
U2 - Beta(0.5, 0.5)	18.4	65.1	14.8	1.2	0	0.5
U2 - Beta(0.7, 0.3)	21.9	59.3	15.6	2.8	0	0.4
U2 - Beta(1, 1)	18.1	65	15.4	1.1	0	0.4
U3 - Beta(0.3, 0.7)	44.2	53.7	1.6	0	0	0.5
U3 - Beta(0.5, 0.5)	20.9	65.7	12.1	0.6	0	0.7
U3 - Beta(0.7, 0.3)	23	59.7	15.5	1.3	0	0.5
U3 - Beta(1, 1)	20.5	68	10.3	0.8	0	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	16.9	11.9	1	0.1	0	-
U2 - Beta(0.5, 0.5)	9.4	15.4	4.4	0.7	0.1	-
U2 - Beta(0.7, 0.3)	9.3	12.7	5.8	2	0.2	-
U2 - Beta(1, 1)	9.3	15.3	4.5	0.7	0.1	-
U3 - Beta(0.3, 0.7)	17.6	11.6	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	10	15.6	3.8	0.5	0	-
U3 - Beta(0.7, 0.3)	9.6	13.5	5.3	1.5	0.1	-
U3 - Beta(1, 1)	10.1	15.9	3.5	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.1	4	0.3	0	0	-
U2 - Beta(0.5, 0.5)	2.1	5.1	1.2	0.2	0	-
U2 - Beta(0.7, 0.3)	2	4.4	1.4	0.6	0.1	-
U2 - Beta(1, 1)	2	5.1	1.2	0.2	0	-
U3 - Beta(0.3, 0.7)	5.4	3.9	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2.3	5.2	1	0.2	0	-
U3 - Beta(0.7, 0.3)	2.1	4.6	1.4	0.4	0	-
U3 - Beta(1, 1)	2.3	5.3	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1126.3
U2 - Beta(0.5, 0.5)						1180.1
U2 - Beta(0.7, 0.3)						1192.1
U2 - Beta(1, 1)						1180.1
U3 - Beta(0.3, 0.7)						1122.1
U3 - Beta(0.5, 0.5)						1173.4
U3 - Beta(0.7, 0.3)						1182.9
U3 - Beta(1, 1)						1173.2
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.1	7.4	11.2	25.9	51.3	0.1
U2 - Beta(0.5, 0.5)	10.3	10.2	8.8	15.7	54.9	0.1
U2 - Beta(0.7, 0.3)	11.6	12	12.2	14.6	49.5	0.1
U2 - Beta(1, 1)	12.5	10.3	9.6	14	53.5	0.1
U3 - Beta(0.3, 0.7)	10.7	9.6	13.8	24.4	41.4	0.1
U3 - Beta(0.5, 0.5)	16.2	13.1	10.2	13.3	47.1	0.1
U3 - Beta(0.7, 0.3)	17.2	14.7	11.6	13.1	43.3	0.1
U3 - Beta(1, 1)	19.8	14.4	9.3	11.2	45.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.9	6.3	8.1	6.9	-
U2 - Beta(0.5, 0.5)	3.4	4.1	5.2	7.1	10.2	-
U2 - Beta(0.7, 0.3)	3.5	4	5.2	6.9	10.4	-
U2 - Beta(1, 1)	3.5	4.1	5.3	6.9	10.1	-
U3 - Beta(0.3, 0.7)	3.9	5.1	6.7	8.1	6.1	-
U3 - Beta(0.5, 0.5)	3.5	4.2	5.4	7.2	9.7	-
U3 - Beta(0.7, 0.3)	3.7	4.3	5.4	6.9	9.7	-
U3 - Beta(1, 1)	3.8	4.4	5.3	7	9.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3	2.5	3.2	2.2	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	3	2.9	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.8	2.9	2.9	-
U2 - Beta(1, 1)	0.2	2.9	1.8	2.9	2.9	-
U3 - Beta(0.3, 0.7)	0.5	3.1	2.7	3.3	1.9	-
U3 - Beta(0.5, 0.5)	0.2	2.9	1.9	3	2.8	-
U3 - Beta(0.7, 0.3)	0.3	3	1.8	2.8	2.8	-
U3 - Beta(1, 1)	0.4	3	1.8	2.9	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1025.9
U2 - Beta(0.5, 0.5)						1060.7
U2 - Beta(0.7, 0.3)						1064.3
U2 - Beta(1, 1)						1060.3
U3 - Beta(0.3, 0.7)						1019.1
U3 - Beta(0.5, 0.5)						1057.3
U3 - Beta(0.7, 0.3)						1057
U3 - Beta(1, 1)						1054.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.5	58.1	17.2	0.9	0	13.3
U2 - Beta(0.5, 0.5)	11.7	56.2	16.7	1.2	0	14.2
U2 - Beta(0.7, 0.3)	12.8	56.2	14.7	2.2	0	14.1
U2 - Beta(1, 1)	12.6	56.6	15.3	1.5	0	14
U3 - Beta(0.3, 0.7)	12.7	60.1	12.9	0.7	0	13.6
U3 - Beta(0.5, 0.5)	13.2	57.8	13.6	0.8	0	14.6
U3 - Beta(0.7, 0.3)	13.2	56.4	14.4	1.8	0	14.2
U3 - Beta(1, 1)	14.3	56.4	14	1	0	14.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	15.2	4.5	0.6	0	-
U2 - Beta(0.5, 0.5)	7.9	12.6	5.9	1.9	0.2	-
U2 - Beta(0.7, 0.3)	7.9	12.2	5.8	2.2	0.3	-
U2 - Beta(1, 1)	7.8	12.7	5.9	1.8	0.2	-
U3 - Beta(0.3, 0.7)	8.5	15.6	3.9	0.5	0	-
U3 - Beta(0.5, 0.5)	8.2	12.8	5.6	1.7	0.2	-
U3 - Beta(0.7, 0.3)	8	12.5	5.8	2	0.2	-
U3 - Beta(1, 1)	8.1	12.9	5.6	1.7	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	6.2	1.5	0.2	0	-
U2 - Beta(0.5, 0.5)	1.7	5.5	1.7	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.7	5.4	1.6	0.8	0.1	-
U2 - Beta(1, 1)	1.6	5.5	1.7	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.1	6.4	1.3	0.2	0	-
U3 - Beta(0.5, 0.5)	1.8	5.6	1.6	0.6	0.1	-
U3 - Beta(0.7, 0.3)	1.8	5.5	1.6	0.7	0.1	-
U3 - Beta(1, 1)	1.8	5.6	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1018.1
U2 - Beta(0.5, 0.5)						1033.5
U2 - Beta(0.7, 0.3)						1039.4
U2 - Beta(1, 1)						1035.1
U3 - Beta(0.3, 0.7)						1015.2
U3 - Beta(0.5, 0.5)						1029.4
U3 - Beta(0.7, 0.3)						1036.7
U3 - Beta(1, 1)						1029.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.1	48.6	30.8	12	0.9	2.6
U2 - Beta(0.5, 0.5)	8	49.8	29.1	9.7	0.9	2.5
U2 - Beta(0.7, 0.3)	9.7	48.1	27.7	11.4	0.7	2.4
U2 - Beta(1, 1)	11.3	46.9	28.1	10.2	1	2.5
U3 - Beta(0.3, 0.7)	7.8	54.5	26	9	0.2	2.5
U3 - Beta(0.5, 0.5)	10.2	53	25.3	8.2	0.7	2.6
U3 - Beta(0.7, 0.3)	11.3	51.5	23.7	9.6	1	2.9
U3 - Beta(1, 1)	14	52.7	20.7	8.8	0.9	2.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.9	13.4	7.8	3.1	0.4	-
U2 - Beta(0.5, 0.5)	4.9	10.6	8.2	4.6	1.4	-
U2 - Beta(0.7, 0.3)	4.9	10.3	8	4.9	1.5	-
U2 - Beta(1, 1)	5	10.7	8.1	4.6	1.4	-
U3 - Beta(0.3, 0.7)	5.3	14.5	7.1	2.5	0.3	-
U3 - Beta(0.5, 0.5)	5.1	11.1	7.8	4.4	1.3	-
U3 - Beta(0.7, 0.3)	5.2	10.9	7.7	4.5	1.3	-
U3 - Beta(1, 1)	5.3	11.4	7.5	4.2	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	6	2.7	1.2	0.2	-
U2 - Beta(0.5, 0.5)	0.7	5	2.6	1.7	0.5	-
U2 - Beta(0.7, 0.3)	0.8	5	2.4	1.8	0.5	-
U2 - Beta(1, 1)	0.8	5.1	2.6	1.7	0.5	-
U3 - Beta(0.3, 0.7)	1	6.4	2.5	1	0.1	-
U3 - Beta(0.5, 0.5)	0.8	5.2	2.5	1.6	0.4	-
U3 - Beta(0.7, 0.3)	0.9	5.2	2.3	1.7	0.4	-
U3 - Beta(1, 1)	1	5.3	2.4	1.6	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1033.2
U2 - Beta(0.5, 0.5)						1052.6
U2 - Beta(0.7, 0.3)						1052.9
U2 - Beta(1, 1)						1052
U3 - Beta(0.3, 0.7)						1028.8
U3 - Beta(0.5, 0.5)						1048.1
U3 - Beta(0.7, 0.3)						1050.3
U3 - Beta(1, 1)						1046.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	56.1	0.1	0	0	0	43.8
U2 - Beta(0.5, 0.5)	52.5	1	0	0	0	46.5
U2 - Beta(0.7, 0.3)	51.1	1.4	0	0	0	47.5
U2 - Beta(1, 1)	52.4	1.1	0	0	0	46.5
U3 - Beta(0.3, 0.7)	56.9	0.1	0	0	0	43
U3 - Beta(0.5, 0.5)	52.9	1	0	0	0	46.1
U3 - Beta(0.7, 0.3)	51.4	1.5	0	0	0	47.1
U3 - Beta(1, 1)	52.5	0.9	0	0	0	46.6
Mean administrations						
U2 - Beta(0.3, 0.7)	22.2	0.5	0	0	0	-
U2 - Beta(0.5, 0.5)	19.8	2.3	0.1	0	0	-
U2 - Beta(0.7, 0.3)	18.5	3.4	0.2	0	0	-
U2 - Beta(1, 1)	19.8	2.3	0.1	0	0	-
U3 - Beta(0.3, 0.7)	22.3	0.5	0	0	0	-
U3 - Beta(0.5, 0.5)	20	2.2	0.1	0	0	-
U3 - Beta(0.7, 0.3)	19.1	2.9	0.2	0	0	-
U3 - Beta(1, 1)	19.9	2.2	0.1	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.7	0.2	0	0	0	-
U2 - Beta(0.5, 0.5)	3.8	0.9	0	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1.2	0.1	0	0	-
U2 - Beta(1, 1)	3.8	0.9	0	0	0	-
U3 - Beta(0.3, 0.7)	4.7	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.8	0.9	0	0	0	-
U3 - Beta(0.7, 0.3)	3.5	1.1	0.1	0	0	-
U3 - Beta(1, 1)	3.8	0.9	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						983.1
U2 - Beta(0.5, 0.5)						968.5
U2 - Beta(0.7, 0.3)						968.5
U2 - Beta(1, 1)						967.8
U3 - Beta(0.3, 0.7)						984.7
U3 - Beta(0.5, 0.5)						968.8
U3 - Beta(0.7, 0.3)						968.2
U3 - Beta(1, 1)						967.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.1	61.2	11.9	1.2	0.1	10.5
U2 - Beta(0.5, 0.5)	15	60.2	12.3	1.3	0	11.2
U2 - Beta(0.7, 0.3)	13.7	60.3	12.2	2.2	0	11.6
U2 - Beta(1, 1)	15.3	58.9	13	1.7	0	11.1
U3 - Beta(0.3, 0.7)	15.5	62.9	9.9	1.2	0	10.5
U3 - Beta(0.5, 0.5)	16.3	61.4	9.9	1.3	0	11.1
U3 - Beta(0.7, 0.3)	17.4	59.3	10	2.2	0	11.1
U3 - Beta(1, 1)	17.9	58.2	10.6	2	0	11.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	15.3	4.6	0.9	0	-
U2 - Beta(0.5, 0.5)	8.4	13.3	5.3	1.8	0.1	-
U2 - Beta(0.7, 0.3)	8.5	12.9	5.4	2.1	0.2	-
U2 - Beta(1, 1)	8.3	13.3	5.4	1.9	0.1	-
U3 - Beta(0.3, 0.7)	8.7	15.6	4	0.7	0	-
U3 - Beta(0.5, 0.5)	8.7	13.4	5	1.7	0.1	-
U3 - Beta(0.7, 0.3)	8.6	13.1	5.2	2	0.1	-
U3 - Beta(1, 1)	8.7	13.4	5.1	1.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.4	1.6	0.3	0	-
U2 - Beta(0.5, 0.5)	2	5.9	1.5	0.7	0	-
U2 - Beta(0.7, 0.3)	2	5.8	1.4	0.7	0.1	-
U2 - Beta(1, 1)	1.9	5.8	1.6	0.7	0	-
U3 - Beta(0.3, 0.7)	2.3	6.5	1.4	0.2	0	-
U3 - Beta(0.5, 0.5)	2.1	5.9	1.5	0.6	0	-
U3 - Beta(0.7, 0.3)	2.1	5.8	1.4	0.7	0	-
U3 - Beta(1, 1)	2.1	5.9	1.5	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.5
U2 - Beta(0.5, 0.5)						1038.6
U2 - Beta(0.7, 0.3)						1042.9
U2 - Beta(1, 1)						1037.1
U3 - Beta(0.3, 0.7)						1022.1
U3 - Beta(0.5, 0.5)						1034.5
U3 - Beta(0.7, 0.3)						1038.7
U3 - Beta(1, 1)						1034.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.6	1	0	0	0	91.4
U2 - Beta(0.5, 0.5)	7.5	0.6	0	0	0	91.9
U2 - Beta(0.7, 0.3)	7.7	0.7	0	0	0	91.6
U2 - Beta(1, 1)	7.4	1.7	0	0	0	90.9
U3 - Beta(0.3, 0.7)	7	0.9	0	0	0	92.1
U3 - Beta(0.5, 0.5)	7.5	1.1	0	0	0	91.4
U3 - Beta(0.7, 0.3)	7.2	0.7	0	0	0	92.1
U3 - Beta(1, 1)	7.5	0.8	0	0	0	91.7
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	3.7	0.5	0	0	-
U2 - Beta(0.5, 0.5)	11.1	3.8	0.5	0	0	-
U2 - Beta(0.7, 0.3)	11.1	3.9	0.5	0.1	0	-
U2 - Beta(1, 1)	11	3.9	0.5	0	0	-
U3 - Beta(0.3, 0.7)	11.3	3.6	0.4	0	0	-
U3 - Beta(0.5, 0.5)	11.1	3.8	0.5	0	0	-
U3 - Beta(0.7, 0.3)	11	3.8	0.5	0.1	0	-
U3 - Beta(1, 1)	11	3.9	0.5	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2.1	0.2	0	0	-
U2 - Beta(0.7, 0.3)	2.9	2.1	0.2	0	0	-
U2 - Beta(1, 1)	2.8	2.1	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3	2	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	2.1	0.2	0	0	-
U3 - Beta(0.7, 0.3)	2.8	2.1	0.2	0	0	-
U3 - Beta(1, 1)	2.8	2.1	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						560.9
U2 - Beta(0.5, 0.5)						557.9
U2 - Beta(0.7, 0.3)						560.6
U2 - Beta(1, 1)						558.4
U3 - Beta(0.3, 0.7)						558.1
U3 - Beta(0.5, 0.5)						557.2
U3 - Beta(0.7, 0.3)						556
U3 - Beta(1, 1)						558.6

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	1.1	8.6	76.9	13.2	0
U2 - Beta(0.5, 0.5)	1.5	1	3	59.1	35.4	0
U2 - Beta(0.7, 0.3)	3.3	2.4	5.7	47.2	41.4	0
U2 - Beta(1, 1)	1.7	1.6	3.2	58.9	34.6	0
U3 - Beta(0.3, 0.7)	0.9	1.6	10.5	76.9	10.1	0
U3 - Beta(0.5, 0.5)	2.7	1.9	2.7	62.6	30.1	0
U3 - Beta(0.7, 0.3)	4.3	3	6	52.6	34.1	0
U3 - Beta(1, 1)	2.8	2.8	3.5	62.5	28.4	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4.1	6.9	13.6	1.7	-
U2 - Beta(0.5, 0.5)	3.1	3.1	3.8	13	6.9	-
U2 - Beta(0.7, 0.3)	3.1	3.1	3.8	11	9	-
U2 - Beta(1, 1)	3.1	3.2	3.9	13.1	6.7	-
U3 - Beta(0.3, 0.7)	3.6	4.2	7.2	13.6	1.4	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	13.6	6.2	-
U3 - Beta(0.7, 0.3)	3.2	3.2	4	11.6	8.1	-
U3 - Beta(1, 1)	3.1	3.2	4	13.8	5.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	2.7	4.7	0.6	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.5	2.1	-
U2 - Beta(0.7, 0.3)	0	2.4	1.2	4	2.5	-
U2 - Beta(1, 1)	0	2.4	1.2	4.5	2	-
U3 - Beta(0.3, 0.7)	0.4	2.7	2.9	4.7	0.5	-
U3 - Beta(0.5, 0.5)	0	2.4	1.2	4.7	1.8	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.2	4.1	2.3	-
U3 - Beta(1, 1)	0.1	2.5	1.3	4.8	1.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1046.4
U2 - Beta(0.5, 0.5)						1090.8
U2 - Beta(0.7, 0.3)						1096.5
U2 - Beta(1, 1)						1089.5
U3 - Beta(0.3, 0.7)						1042.8
U3 - Beta(0.5, 0.5)						1088.3
U3 - Beta(0.7, 0.3)						1091.1
U3 - Beta(1, 1)						1086.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	23.2	39.3	33.9	2.1	0.2	1.3
U2 - Beta(0.5, 0.5)	28.2	33.2	33	4.1	0.1	1.4
U2 - Beta(0.7, 0.3)	29.8	33.7	30	5	0	1.5
U2 - Beta(1, 1)	28.7	32.3	33.9	3.8	0	1.3
U3 - Beta(0.3, 0.7)	32.1	41.2	24	1.4	0	1.3
U3 - Beta(0.5, 0.5)	32.2	34.2	29.5	2.8	0	1.3
U3 - Beta(0.7, 0.3)	32.1	33.6	28.5	4.2	0.1	1.5
U3 - Beta(1, 1)	33.1	33.6	28.2	3.5	0	1.6
Mean administrations						
U2 - Beta(0.3, 0.7)	10.7	11.3	7	0.8	0	-
U2 - Beta(0.5, 0.5)	8.8	9.9	8.5	2.4	0.2	-
U2 - Beta(0.7, 0.3)	9.2	9.7	7.7	2.9	0.4	-
U2 - Beta(1, 1)	8.8	9.9	8.6	2.4	0.2	-
U3 - Beta(0.3, 0.7)	12.3	11.7	5.4	0.5	0	-
U3 - Beta(0.5, 0.5)	9.4	10	8.2	2.1	0.2	-
U3 - Beta(0.7, 0.3)	9.4	9.7	7.7	2.7	0.3	-
U3 - Beta(1, 1)	9.6	10.1	7.9	2.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	4.7	2.2	0.3	0	-
U2 - Beta(0.5, 0.5)	2.2	4.3	2.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.4	4.3	2	0.9	0.1	-
U2 - Beta(1, 1)	2.2	4.3	2.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	3.8	4.9	1.7	0.2	0	-
U3 - Beta(0.5, 0.5)	2.5	4.3	2.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	4.2	2.1	0.9	0.1	-
U3 - Beta(1, 1)	2.5	4.4	2.2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1073.6
U2 - Beta(0.5, 0.5)						1109.8
U2 - Beta(0.7, 0.3)						1112.8
U2 - Beta(1, 1)						1110.7
U3 - Beta(0.3, 0.7)						1062.1
U3 - Beta(0.5, 0.5)						1103.1
U3 - Beta(0.7, 0.3)						1108.4
U3 - Beta(1, 1)						1100.2

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.6	32.6	37.5	1.7	0	11.6
U2 - Beta(0.5, 0.5)	20.6	33.2	28.6	5.2	0.3	12.1
U2 - Beta(0.7, 0.3)	20.2	34.8	28.2	4.4	0.1	12.3
U2 - Beta(1, 1)	19.7	33	29.2	5.9	0.2	12
U3 - Beta(0.3, 0.7)	20.4	33.9	31.1	2	0	12.6
U3 - Beta(0.5, 0.5)	23.7	33.1	26.7	4.1	0.2	12.2
U3 - Beta(0.7, 0.3)	24.4	32.8	24.7	5.1	0.2	12.8
U3 - Beta(1, 1)	23.6	34.5	24.4	4.8	0.3	12.4
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.2	8	1.2	0	-
U2 - Beta(0.5, 0.5)	8.7	9.2	7.7	2.7	0.3	-
U2 - Beta(0.7, 0.3)	9	9.2	7.1	2.9	0.4	-
U2 - Beta(1, 1)	8.5	9.4	7.7	2.7	0.3	-
U3 - Beta(0.3, 0.7)	9.8	10.7	7	0.9	0	-
U3 - Beta(0.5, 0.5)	9.4	9.4	7.5	2.4	0.2	-
U3 - Beta(0.7, 0.3)	9.1	9.2	7.1	2.8	0.2	-
U3 - Beta(1, 1)	9	9.5	7.3	2.4	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	4.6	2.6	0.5	0	-
U2 - Beta(0.5, 0.5)	2.2	4.3	2.2	1	0.1	-
U2 - Beta(0.7, 0.3)	2.3	4.4	1.9	1	0.1	-
U2 - Beta(1, 1)	2	4.4	2.2	1	0.1	-
U3 - Beta(0.3, 0.7)	2.7	4.8	2.3	0.4	0	-
U3 - Beta(0.5, 0.5)	2.3	4.4	2.1	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.3	4.4	2	1	0.1	-
U3 - Beta(1, 1)	2.3	4.5	2	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1006.6
U2 - Beta(0.5, 0.5)						1030
U2 - Beta(0.7, 0.3)						1032
U2 - Beta(1, 1)						1030.2
U3 - Beta(0.3, 0.7)						996.2
U3 - Beta(0.5, 0.5)						1024.5
U3 - Beta(0.7, 0.3)						1024.1
U3 - Beta(1, 1)						1021.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11	27.6	50.9	5.4	0.4	4.7
U2 - Beta(0.5, 0.5)	14.4	25.7	45.5	9.3	0.4	4.7
U2 - Beta(0.7, 0.3)	14.5	25.4	44.6	10.1	0.3	5.1
U2 - Beta(1, 1)	14.7	23	46.8	10.7	0.3	4.5
U3 - Beta(0.3, 0.7)	15.2	30.7	44.6	4.1	0.2	5.2
U3 - Beta(0.5, 0.5)	17.4	27.7	42.3	7.5	0.2	4.9
U3 - Beta(0.7, 0.3)	19.6	25.4	41.3	9	0.2	4.5
U3 - Beta(1, 1)	18.2	25.9	43.2	7.8	0.2	4.7
Mean administrations						
U2 - Beta(0.3, 0.7)	7.1	10.3	10.2	1.6	0.1	-
U2 - Beta(0.5, 0.5)	6.7	8.6	10	3.5	0.6	-
U2 - Beta(0.7, 0.3)	6.8	8.5	9.1	4.1	0.9	-
U2 - Beta(1, 1)	6.5	8.6	10.2	3.5	0.6	-
U3 - Beta(0.3, 0.7)	8.1	10.8	9.3	1.2	0	-
U3 - Beta(0.5, 0.5)	7	8.8	9.8	3.2	0.5	-
U3 - Beta(0.7, 0.3)	7	8.8	9.2	3.9	0.6	-
U3 - Beta(1, 1)	7.1	8.9	9.7	3.2	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	4.6	3.3	0.6	0	-
U2 - Beta(0.5, 0.5)	1.4	4.1	2.9	1.2	0.2	-
U2 - Beta(0.7, 0.3)	1.5	4.1	2.6	1.3	0.2	-
U2 - Beta(1, 1)	1.4	4.1	3	1.2	0.2	-
U3 - Beta(0.3, 0.7)	2.2	4.8	3	0.4	0	-
U3 - Beta(0.5, 0.5)	1.6	4.2	2.9	1.1	0.2	-
U3 - Beta(0.7, 0.3)	1.6	4.2	2.6	1.3	0.2	-
U3 - Beta(1, 1)	1.6	4.2	2.8	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1047
U2 - Beta(0.5, 0.5)						1077.5
U2 - Beta(0.7, 0.3)						1079.7
U2 - Beta(1, 1)						1076.7
U3 - Beta(0.3, 0.7)						1038.3
U3 - Beta(0.5, 0.5)						1071.9
U3 - Beta(0.7, 0.3)						1078.6
U3 - Beta(1, 1)						1070.8
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.8	41.9	4.9	0.1	0	44.3
U2 - Beta(0.5, 0.5)	11.2	39.2	5.4	0.1	0	44.1
U2 - Beta(0.7, 0.3)	11.5	38.2	6	0	0	44.3
U2 - Beta(1, 1)	11.8	37.6	5.7	0.1	0	44.8
U3 - Beta(0.3, 0.7)	9.8	41.7	3.7	0.1	0	44.7
U3 - Beta(0.5, 0.5)	11.6	38.7	5.1	0	0	44.6
U3 - Beta(0.7, 0.3)	11.9	39.3	4.7	0	0	44.1
U3 - Beta(1, 1)	12.3	37.7	5.2	0	0	44.8
Mean administrations						
U2 - Beta(0.3, 0.7)	9.5	12.2	2.2	0.3	0	-
U2 - Beta(0.5, 0.5)	9.4	10.7	3.3	0.8	0.1	-
U2 - Beta(0.7, 0.3)	9.5	10.3	3.4	1.1	0.1	-
U2 - Beta(1, 1)	9.4	10.7	3.3	0.8	0.1	-
U3 - Beta(0.3, 0.7)	9.7	12.3	2	0.2	0	-
U3 - Beta(0.5, 0.5)	9.6	10.8	3.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	9.6	10.5	3.3	1	0.1	-
U3 - Beta(1, 1)	9.7	10.7	3.1	0.8	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	5.2	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	2.3	4.7	1	0.3	0	-
U2 - Beta(0.7, 0.3)	2.3	4.6	1	0.4	0	-
U2 - Beta(1, 1)	2.2	4.6	1	0.3	0	-
U3 - Beta(0.3, 0.7)	2.4	5.2	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	2.3	4.7	0.9	0.3	0	-
U3 - Beta(0.7, 0.3)	2.3	4.6	1	0.4	0	-
U3 - Beta(1, 1)	2.4	4.7	0.9	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						869.5
U2 - Beta(0.5, 0.5)						880.4
U2 - Beta(0.7, 0.3)						888.4
U2 - Beta(1, 1)						882.7
U3 - Beta(0.3, 0.7)						867.4
U3 - Beta(0.5, 0.5)						883.8
U3 - Beta(0.7, 0.3)						886
U3 - Beta(1, 1)						879.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	81.1	15.7	0.7	0	0	2.5
U2 - Beta(0.5, 0.5)	63.7	31.4	2.1	0.1	0	2.7
U2 - Beta(0.7, 0.3)	66.3	28.5	2.8	0.2	0	2.2
U2 - Beta(1, 1)	64.5	30.3	2.6	0.1	0	2.5
U3 - Beta(0.3, 0.7)	82.4	15	0.2	0	0	2.4
U3 - Beta(0.5, 0.5)	67.5	28.1	1.8	0.1	0	2.5
U3 - Beta(0.7, 0.3)	67.2	27.6	2.6	0.2	0	2.4
U3 - Beta(1, 1)	67.8	28.3	1.5	0.1	0	2.3
Mean administrations						
U2 - Beta(0.3, 0.7)	24.9	4.2	0.4	0	0	-
U2 - Beta(0.5, 0.5)	18.1	9	2	0.4	0	-
U2 - Beta(0.7, 0.3)	17.6	8.8	2.5	0.7	0	-
U2 - Beta(1, 1)	18	9	2.1	0.4	0	-
U3 - Beta(0.3, 0.7)	25.2	4.1	0.2	0	0	-
U3 - Beta(0.5, 0.5)	19	8.6	1.6	0.3	0	-
U3 - Beta(0.7, 0.3)	17.8	8.9	2.3	0.5	0	-
U3 - Beta(1, 1)	19	8.6	1.6	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7	1.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.4	3.2	0.6	0.1	0	-
U2 - Beta(0.7, 0.3)	4.3	3.1	0.7	0.2	0	-
U2 - Beta(1, 1)	4.4	3.2	0.6	0.1	0	-
U3 - Beta(0.3, 0.7)	7.1	1.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.8	3.1	0.5	0.1	0	-
U3 - Beta(0.7, 0.3)	4.3	3.2	0.7	0.2	0	-
U3 - Beta(1, 1)	4.8	3.1	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1147.6
U2 - Beta(0.5, 0.5)						1166.3
U2 - Beta(0.7, 0.3)						1178.1
U2 - Beta(1, 1)						1167.5
U3 - Beta(0.3, 0.7)						1148.3
U3 - Beta(0.5, 0.5)						1164.5
U3 - Beta(0.7, 0.3)						1171.8
U3 - Beta(1, 1)						1164.9
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.6	11.2	19.1	54.3	5.4	0.4
U2 - Beta(0.5, 0.5)	16.3	10.8	15.6	46	10.9	0.4
U2 - Beta(0.7, 0.3)	18.7	14.9	14.3	40.5	11.2	0.4
U2 - Beta(1, 1)	17.1	13.6	13.1	44.7	11.1	0.4
U3 - Beta(0.3, 0.7)	16	15.1	18.8	46.9	2.8	0.4
U3 - Beta(0.5, 0.5)	23.5	14.2	12.9	39.7	9.2	0.5
U3 - Beta(0.7, 0.3)	25	17.9	13.2	34.4	9.2	0.3
U3 - Beta(1, 1)	25.1	16.7	13	36.6	8.2	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	6.7	7.6	8.8	1.1	-
U2 - Beta(0.5, 0.5)	4.4	5.2	6.4	10.1	3.8	-
U2 - Beta(0.7, 0.3)	4.7	5.1	6.4	9.4	4.3	-
U2 - Beta(1, 1)	4.4	5.3	6.4	10.1	3.7	-
U3 - Beta(0.3, 0.7)	6.5	7.4	7.2	8.1	0.8	-
U3 - Beta(0.5, 0.5)	4.7	5.5	6.4	10	3.4	-
U3 - Beta(0.7, 0.3)	5	5.5	6.2	9.3	3.8	-
U3 - Beta(1, 1)	4.9	5.5	6.3	10	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	3.5	2.9	3	0.4	-
U2 - Beta(0.5, 0.5)	0.6	3.1	2	3.3	1.1	-
U2 - Beta(0.7, 0.3)	0.7	3.1	2	3.2	1.2	-
U2 - Beta(1, 1)	0.6	3.1	2.1	3.3	1.1	-
U3 - Beta(0.3, 0.7)	1.6	3.8	2.8	2.7	0.3	-
U3 - Beta(0.5, 0.5)	0.7	3.2	2	3.3	1	-
U3 - Beta(0.7, 0.3)	0.9	3.3	1.9	3.1	1.1	-
U3 - Beta(1, 1)	0.8	3.2	2	3.3	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039.2
U2 - Beta(0.5, 0.5)						1085.8
U2 - Beta(0.7, 0.3)						1088.1
U2 - Beta(1, 1)						1085.8
U3 - Beta(0.3, 0.7)						1030.8
U3 - Beta(0.5, 0.5)						1081.6
U3 - Beta(0.7, 0.3)						1081.6
U3 - Beta(1, 1)						1079.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.1	76.3	15.3	2.1	0.2	0
U2 - Beta(0.5, 0.5)	2.9	38.1	31.3	23.2	4.5	0
U2 - Beta(0.7, 0.3)	6.1	34.6	25.6	25.5	8.2	0
U2 - Beta(1, 1)	3.3	37.7	30.8	23.3	4.9	0
U3 - Beta(0.3, 0.7)	9.1	77.7	11.9	1.3	0	0
U3 - Beta(0.5, 0.5)	4.8	44.8	28.3	20	2.1	0
U3 - Beta(0.7, 0.3)	7.1	37.9	25.2	25.9	3.9	0
U3 - Beta(1, 1)	5.2	44	29.9	18.8	2.1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	18.5	3	0.3	0	-
U2 - Beta(0.5, 0.5)	3.8	12.5	7.6	4.8	1.2	-
U2 - Beta(0.7, 0.3)	3.6	6.7	7.7	8.3	3.6	-
U2 - Beta(1, 1)	3.8	12.4	7.8	4.7	1.2	-
U3 - Beta(0.3, 0.7)	8.6	18.8	2.4	0.3	0	-
U3 - Beta(0.5, 0.5)	4	13.6	7.3	4.3	0.8	-
U3 - Beta(0.7, 0.3)	3.8	8.8	7.8	7.2	2.4	-
U3 - Beta(1, 1)	4	13.4	7.5	4.3	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	6.7	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	0.3	4.9	2.3	1.4	0.4	-
U2 - Beta(0.7, 0.3)	0.2	3.1	2.2	2.2	1	-
U2 - Beta(1, 1)	0.3	4.8	2.3	1.4	0.4	-
U3 - Beta(0.3, 0.7)	2.4	6.8	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	0.3	5.2	2.2	1.3	0.3	-
U3 - Beta(0.7, 0.3)	0.3	3.7	2.2	2	0.7	-
U3 - Beta(1, 1)	0.3	5.2	2.3	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1103.5
U2 - Beta(0.5, 0.5)						1145.8
U2 - Beta(0.7, 0.3)						1171.2
U2 - Beta(1, 1)						1145
U3 - Beta(0.3, 0.7)						1101.7
U3 - Beta(0.5, 0.5)						1140.5
U3 - Beta(0.7, 0.3)						1159.6
U3 - Beta(1, 1)						1139.8

Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.6	8.1	45.9	33.9	8.4	0.1
U2 - Beta(0.5, 0.5)	11.4	14	35	29.1	10.4	0.1
U2 - Beta(0.7, 0.3)	13.1	15.6	33.3	27.9	10	0.1
U2 - Beta(1, 1)	13.6	15	32.7	29.1	9.5	0.1
U3 - Beta(0.3, 0.7)	6.8	11.8	46.9	29.1	5.3	0.1
U3 - Beta(0.5, 0.5)	16.5	18.3	33.6	25.8	5.7	0.1
U3 - Beta(0.7, 0.3)	18.5	18.4	33.2	24.6	5.2	0.1
U3 - Beta(1, 1)	20.4	18.3	31.2	24.8	5.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	6.1	11.6	6.4	1.6	-
U2 - Beta(0.5, 0.5)	3.5	4.3	8.7	8.7	4.7	-
U2 - Beta(0.7, 0.3)	3.6	4.4	8.6	8.3	5	-
U2 - Beta(1, 1)	3.5	4.5	8.8	8.6	4.6	-
U3 - Beta(0.3, 0.7)	4.4	6.6	11.8	5.9	1.2	-
U3 - Beta(0.5, 0.5)	3.6	4.6	9.1	8.6	4	-
U3 - Beta(0.7, 0.3)	3.7	4.8	8.8	8.4	4.3	-
U3 - Beta(1, 1)	3.8	4.9	9	8.5	3.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	4.4	2.3	0.6	-
U2 - Beta(0.5, 0.5)	0.2	2.9	2.9	3.3	1.5	-
U2 - Beta(0.7, 0.3)	0.3	2.9	2.8	3.2	1.6	-
U2 - Beta(1, 1)	0.2	2.9	2.9	3.3	1.5	-
U3 - Beta(0.3, 0.7)	0.7	3.6	4.4	2.2	0.4	-
U3 - Beta(0.5, 0.5)	0.3	3	3	3.2	1.3	-
U3 - Beta(0.7, 0.3)	0.3	3.1	2.9	3.2	1.4	-
U3 - Beta(1, 1)	0.4	3.1	2.9	3.2	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1027
U2 - Beta(0.5, 0.5)						1055.9
U2 - Beta(0.7, 0.3)						1057.1
U2 - Beta(1, 1)						1054.8
U3 - Beta(0.3, 0.7)						1023
U3 - Beta(0.5, 0.5)						1052.1
U3 - Beta(0.7, 0.3)						1053.9
U3 - Beta(1, 1)						1051.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.4	3.9	15.5	63.3	13.8	0.1
U2 - Beta(0.5, 0.5)	8.8	5.9	11.6	47.3	26.3	0.1
U2 - Beta(0.7, 0.3)	9.3	6.8	13	44.1	26.7	0.1
U2 - Beta(1, 1)	9.2	6.1	11.1	46.6	26.9	0.1
U3 - Beta(0.3, 0.7)	6.4	4.7	17.9	61.5	9.4	0.1
U3 - Beta(0.5, 0.5)	12.2	8.2	12	46.1	21.4	0.1
U3 - Beta(0.7, 0.3)	14.2	8.4	13.1	42.3	21.9	0.1
U3 - Beta(1, 1)	13.8	8.7	11.8	44.3	21.3	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.4	7.9	11.7	2.2	-
U2 - Beta(0.5, 0.5)	3.4	3.7	6	10.7	6.2	-
U2 - Beta(0.7, 0.3)	3.4	3.8	6	9.9	7	-
U2 - Beta(1, 1)	3.4	3.8	6	10.6	6.1	-
U3 - Beta(0.3, 0.7)	3.8	4.5	8.3	11.6	1.8	-
U3 - Beta(0.5, 0.5)	3.5	3.8	6.2	10.9	5.7	-
U3 - Beta(0.7, 0.3)	3.6	3.9	6	10.2	6.3	-
U3 - Beta(1, 1)	3.7	4	6	10.8	5.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.9	4	0.7	-
U2 - Beta(0.5, 0.5)	0.2	2.7	1.8	3.8	1.7	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.8	3.6	1.8	-
U2 - Beta(1, 1)	0.2	2.7	1.8	3.8	1.7	-
U3 - Beta(0.3, 0.7)	0.4	2.9	3.1	3.9	0.5	-
U3 - Beta(0.5, 0.5)	0.2	2.8	1.9	3.8	1.5	-
U3 - Beta(0.7, 0.3)	0.3	2.8	1.8	3.6	1.7	-
U3 - Beta(1, 1)	0.3	2.8	1.8	3.8	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1056.3
U2 - Beta(0.5, 0.5)						1090.3
U2 - Beta(0.7, 0.3)						1093.4
U2 - Beta(1, 1)						1089.3
U3 - Beta(0.3, 0.7)						1051.1
U3 - Beta(0.5, 0.5)						1087
U3 - Beta(0.7, 0.3)						1089.4
U3 - Beta(1, 1)						1085.2
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	11	56.6	17.1	2.4	4.8
U2 - Beta(0.5, 0.5)	12.7	13.2	47.5	19.2	3.1	4.3
U2 - Beta(0.7, 0.3)	13.6	13.9	45.8	19.4	3.5	3.8
U2 - Beta(1, 1)	14	12	47.7	18.9	3.3	4.1
U3 - Beta(0.3, 0.7)	10	15.2	52.4	16	1.5	4.9
U3 - Beta(0.5, 0.5)	15.8	16	44.1	17.9	2.2	4
U3 - Beta(0.7, 0.3)	18.1	15.9	41	18.3	2.9	3.8
U3 - Beta(1, 1)	18.7	15.8	41	18.8	2.1	3.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	6.9	12.6	4.3	0.6	-
U2 - Beta(0.5, 0.5)	5.1	6.7	9.9	5.9	2.1	-
U2 - Beta(0.7, 0.3)	5.1	6.5	9.7	6	2.3	-
U2 - Beta(1, 1)	5.2	6.6	10	5.9	2.1	-
U3 - Beta(0.3, 0.7)	5.6	7.3	12.3	3.9	0.5	-
U3 - Beta(0.5, 0.5)	5.5	6.8	9.9	5.8	1.7	-
U3 - Beta(0.7, 0.3)	5.5	6.7	9.4	5.9	2	-
U3 - Beta(1, 1)	5.8	6.9	9.6	5.6	1.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.6	4.3	1.6	0.2	-
U2 - Beta(0.5, 0.5)	0.9	3.7	3.1	2.1	0.7	-
U2 - Beta(0.7, 0.3)	0.9	3.6	3	2.1	0.7	-
U2 - Beta(1, 1)	0.9	3.6	3.1	2.1	0.7	-
U3 - Beta(0.3, 0.7)	1.2	3.8	4.2	1.4	0.2	-
U3 - Beta(0.5, 0.5)	1.1	3.8	3.1	2	0.6	-
U3 - Beta(0.7, 0.3)	1.1	3.7	2.9	2.1	0.7	-
U3 - Beta(1, 1)	1.3	3.8	3	2	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1040.4
U2 - Beta(0.5, 0.5)						1054
U2 - Beta(0.7, 0.3)						1056.9
U2 - Beta(1, 1)						1053.7
U3 - Beta(0.3, 0.7)						1035.6
U3 - Beta(0.5, 0.5)						1050.1
U3 - Beta(0.7, 0.3)						1050.4
U3 - Beta(1, 1)						1049

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8	72.7	17.9	1.3	0	0.1
U2 - Beta(0.5, 0.5)	7.3	43.9	35.2	11.5	2	0.1
U2 - Beta(0.7, 0.3)	10	41.2	30.5	15.8	2.4	0.1
U2 - Beta(1, 1)	8.3	42	35	12.3	2.3	0.1
U3 - Beta(0.3, 0.7)	11.1	74.7	13.3	0.8	0	0.1
U3 - Beta(0.5, 0.5)	11	46.5	30.8	9.7	1.9	0.1
U3 - Beta(0.7, 0.3)	12.3	47.7	28.2	10.1	1.6	0.1
U3 - Beta(1, 1)	11.5	49.6	29.5	8	1.3	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	17.8	3.4	0.4	0	-
U2 - Beta(0.5, 0.5)	4.4	12.3	8.6	3.7	0.9	-
U2 - Beta(0.7, 0.3)	4.3	9.4	8.8	5.6	1.9	-
U2 - Beta(1, 1)	4.4	12.2	8.5	3.8	1	-
U3 - Beta(0.3, 0.7)	8.8	18.1	2.7	0.3	0	-
U3 - Beta(0.5, 0.5)	4.6	13.2	8	3.4	0.8	-
U3 - Beta(0.7, 0.3)	4.7	11	8.4	4.7	1.2	-
U3 - Beta(1, 1)	4.7	13.4	7.9	3.3	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	6.6	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	0.5	4.8	2.4	1.2	0.3	-
U2 - Beta(0.7, 0.3)	0.5	3.9	2.3	1.6	0.6	-
U2 - Beta(1, 1)	0.5	4.8	2.4	1.2	0.3	-
U3 - Beta(0.3, 0.7)	2.6	6.7	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	0.6	5.2	2.2	1	0.2	-
U3 - Beta(0.7, 0.3)	0.7	4.4	2.2	1.4	0.3	-
U3 - Beta(1, 1)	0.6	5.3	2.2	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1093.7
U2 - Beta(0.5, 0.5)						1149
U2 - Beta(0.7, 0.3)						1165.7
U2 - Beta(1, 1)						1148.7
U3 - Beta(0.3, 0.7)						1090.4
U3 - Beta(0.5, 0.5)						1142.2
U3 - Beta(0.7, 0.3)						1154.1
U3 - Beta(1, 1)						1140.8

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.7	40.1	16.6	5.6	0.9	25.1
U2 - Beta(0.5, 0.5)	13.4	41.4	15.4	5.1	1	23.7
U2 - Beta(0.7, 0.3)	13.8	40.7	15.6	5.1	1.7	23.1
U2 - Beta(1, 1)	14.1	40.6	15.2	4.8	1.6	23.7
U3 - Beta(0.3, 0.7)	14.4	41.6	13.6	4	0.7	25.7
U3 - Beta(0.5, 0.5)	16.8	42.4	13.5	4	0.5	22.8
U3 - Beta(0.7, 0.3)	16.2	42	13.3	4.5	0.7	23.3
U3 - Beta(1, 1)	18.4	39.9	12.3	4.7	0.5	24.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	11.6	5	1.9	0.3	-
U2 - Beta(0.5, 0.5)	8.1	10.2	5.3	2.9	0.8	-
U2 - Beta(0.7, 0.3)	8.1	10	5.2	3.1	1	-
U2 - Beta(1, 1)	8.1	10.3	5.3	2.9	0.8	-
U3 - Beta(0.3, 0.7)	8.6	11.9	4.6	1.7	0.2	-
U3 - Beta(0.5, 0.5)	8.4	10.3	5.1	2.7	0.7	-
U3 - Beta(0.7, 0.3)	8.4	10.2	5	2.9	0.8	-
U3 - Beta(1, 1)	8.5	10.3	4.9	2.8	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	5.4	1.8	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.9	4.9	1.7	1.1	0.3	-
U2 - Beta(0.7, 0.3)	1.8	4.9	1.6	1.2	0.3	-
U2 - Beta(1, 1)	1.8	5	1.7	1.1	0.3	-
U3 - Beta(0.3, 0.7)	2.1	5.4	1.6	0.7	0.1	-
U3 - Beta(0.5, 0.5)	2	5	1.6	1	0.2	-
U3 - Beta(0.7, 0.3)	2	5	1.5	1.1	0.2	-
U3 - Beta(1, 1)	2	5	1.5	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						943.5
U2 - Beta(0.5, 0.5)						958.5
U2 - Beta(0.7, 0.3)						963.5
U2 - Beta(1, 1)						960.1
U3 - Beta(0.3, 0.7)						937.2
U3 - Beta(0.5, 0.5)						957
U3 - Beta(0.7, 0.3)						958.3
U3 - Beta(1, 1)						953.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.6	49.2	15.7	2.3	0.2	20
U2 - Beta(0.5, 0.5)	13.4	50.2	13.7	2.5	0.4	19.8
U2 - Beta(0.7, 0.3)	13.6	49.8	14	2.8	0.5	19.3
U2 - Beta(1, 1)	14.3	47.5	14.7	2.9	0.6	20
U3 - Beta(0.3, 0.7)	15.1	51.3	11.7	1.7	0.2	20
U3 - Beta(0.5, 0.5)	16.1	50.3	11.6	2.4	0.2	19.4
U3 - Beta(0.7, 0.3)	17.8	48.1	12.4	2.4	0.2	19.1
U3 - Beta(1, 1)	17.5	48.6	11.5	2.5	0.3	19.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	12.9	5.1	1.6	0.2	-
U2 - Beta(0.5, 0.5)	7.9	11.6	5.4	2.5	0.7	-
U2 - Beta(0.7, 0.3)	7.9	11.2	5.4	2.7	0.7	-
U2 - Beta(1, 1)	7.9	11.5	5.4	2.5	0.7	-
U3 - Beta(0.3, 0.7)	8.4	13.4	4.6	1.4	0.2	-
U3 - Beta(0.5, 0.5)	8.3	11.5	5.2	2.4	0.6	-
U3 - Beta(0.7, 0.3)	8.3	11.5	5.2	2.5	0.6	-
U3 - Beta(1, 1)	8.3	11.6	5.1	2.4	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5.9	1.8	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.8	5.5	1.7	1	0.2	-
U2 - Beta(0.7, 0.3)	1.8	5.4	1.7	1.1	0.2	-
U2 - Beta(1, 1)	1.8	5.4	1.7	1	0.2	-
U3 - Beta(0.3, 0.7)	2.1	6	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2	5.5	1.6	0.9	0.2	-
U3 - Beta(0.7, 0.3)	2	5.5	1.6	1	0.2	-
U3 - Beta(1, 1)	2	5.5	1.6	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						966.8
U2 - Beta(0.5, 0.5)						979.6
U2 - Beta(0.7, 0.3)						979.7
U2 - Beta(1, 1)						977.7
U3 - Beta(0.3, 0.7)						964.6
U3 - Beta(0.5, 0.5)						975.8
U3 - Beta(0.7, 0.3)						976.9
U3 - Beta(1, 1)						975.1